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Recovery-Aware Physical Event Attribution From Sparse PMUs

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ABSTRACT Sparse PMUs can reveal a disturbance without identifying its physical source, especially when a new intervention begins during recovery from an earlier one. We study this ambiguity through competing intervention-conditioned network trajectories. A frozen differential-algebraic model predicts voltage, terminal current, frequency and ROCOF at eight PMUs. A causal history bank tests load and line hypotheses over a ten-second retrospective window, integrates load severity, retains the originating cause after restoration, and combines fixed network-model hypotheses. Its source masses M and increments distinguish conditional evidence from model reweighting. Physical restoration changes the operator without erasing the state, so recovery remains a candidate explanation for subsequent measurements. Separate experiments expose the information limit: accurate hidden-voltage recovery does not imply source separation, and weak-pair exact-support Top-1 in the matched batch study is 24.4%, versus 100.0% for finite pairs. In continuous-stream development tests, the bank obtains 53/54 timely family/source decisions with 4 unmatched alarms. Nominal gains one timely decision at a higher alarm burden; stricter admission loses a real intervention. Confirmatory performance remains pending completion of the frozen campaign. The demonstrated scope is a known load/line catalog under stated network uncertainty, without a universal optimality or all-family claim.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

AN unmeasured load changes. Several PMUs respond, a detector raises an alarm, and the network keeps oscillating after the load is restored. A second intervention then begins during that recovery. The operational question is no longer whether one signal exceeds a threshold: which physical intervention explains the reports observed so far, where did it originate, and when is that explanation distinguishable from the alternatives?

Sparse PMUs make the question difficult because a source need not have a sensor at its own bus. Its voltage, current and frequency effects propagate through the network [1], [2]. Uncertain operating context and network parameters can also change those effects. Moreover, physical OFF is not electrical recovery. A method that resets at restoration can treat the remaining response as another event or lose a real event that begins before the oscillation decays.

Strong existing methods solve parts of this problem. Dynamic state estimators combine power-system equations and PMU histories [3]; supervised classifiers learn sparse-PMU event/source labels [4], [5]; physical outage detectors compare network hypotheses sequentially [6], [7]; and high-fidelity fault trajectories already support sparse-PMU diagnosis [8]. The question here is how to retain an explanation through restoration and compare its recovery with a later intervention. A low state error does not prove source identity, and a score on represented assets does not prove source transfer.

Our approach uses the physical intervention as the object of inference. A load or line candidate changes the network's DAE operator. Starting from an estimated past state, that operator predicts what all eight PMUs should report. Conditional likelihoods compare those responses with the observed history while retaining amplitude, noise and network-model uncertainty. The source/family masses M summarize the competing explanations; their increments separate conditional evidence evolution from reweighting physical models. Retained recovery states compete against new events rather than being discarded at OFF.

The paper makes three scoped contributions:

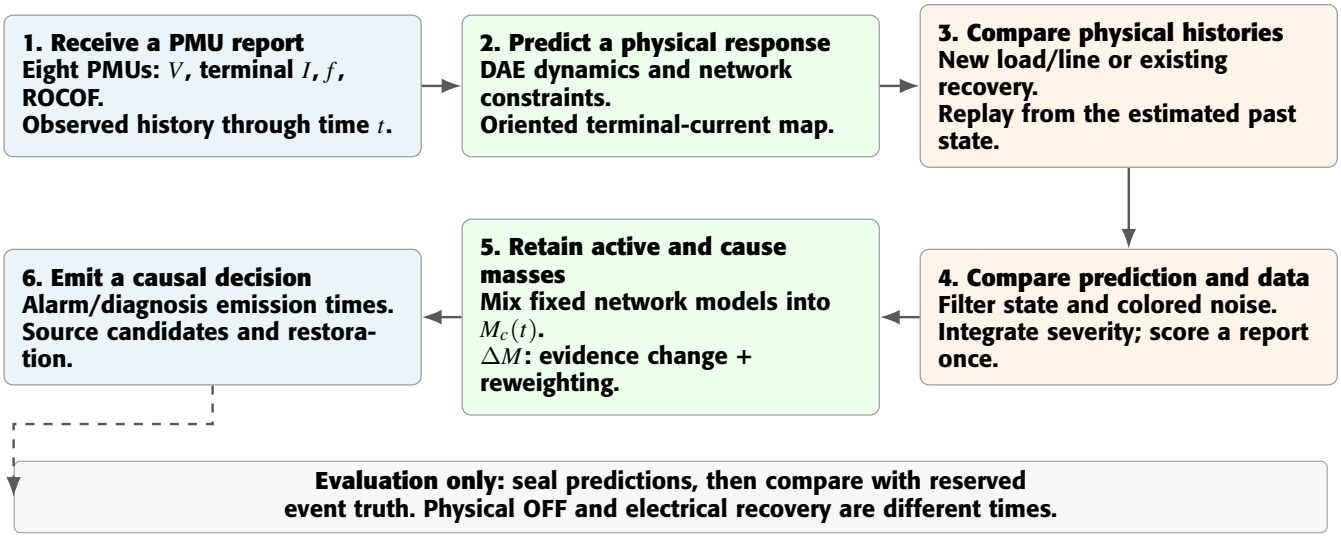


FIGURE 1. **The complete scientific argument. Physics maps an intervention at an unmeasured source into predicted PMU trajectories; causal evidence chooses among those explanations. Event truth enters only evaluation. The figure describes the implemented restricted load/line component, not an already validated all-family joint estimator.**

1) **Recovery-aware physical attribution:** an executable causal load/line history bank starts from estimated past states, retains the originating cause after physical OFF, and compares recovery with new-intervention histories under fixed network hypotheses. We specify its evidence and publication rules.
2) **A bounded source-resolution analysis:** applying established observability and diagnosability ideas, the matched load-support study distinguishes accurate state reconstruction from reliable source identity and exposes weak-source ambiguity. This is a separate component test, not a causal optimality theorem.
3) **A reproducible operational test:** the nominal/mixture comparison and a failed admission ablation expose the trade-off between timely source attribution and extra alarms. Separate state, batch and stream contracts retain their own denominators. R11 and nominal remain frozen co-references for confirmation.

The methodological contribution is the specified competition between intervention and recovery histories under the declared sparse-PMU contract. Physical templates, retrospective onset search, standard Bayesian updates and the array name are established or elementary ingredients. We claim neither their first use nor a globally optimal filter. In completed development evidence, the five-model bank obtains 53/54 timely family/source decisions with 4 extras; nominal retains one more timely event at a higher alarm burden. These tests support a restricted capability, not numerical superiority over every cited method. Final performance claims require the separate frozen confirmation.

II. CLOSEST METHODS AND THE REMAINING QUESTION

Dynamic state estimation supplies the state and uncertainty machinery used to predict a PMU history. Unknown-input and DAE estimators address incomplete models and exogenous inputs [3], [9], [10]. Their task is not automatically an event-family/source decision. A recovered bus-voltage field can remain compatible with competing physical interventions.

Model-based source inference is the closest methodological precedent. Phasor angles support outage hypotheses [11]; uncertain pre-event conditions can be marginalized with outage identity [12]; and sequential likelihoods detect line changes under stochastic operation [6], [7]. Transient dynamics and unknown onset are therefore established ingredients, not contributions of this paper.

Warner *et al.* provide a particularly close precedent [8]. Their simulated voltage/current fault trajectories are compared with a steady-state reference over a moving onset window; Algorithm 1 stops and returns candidate lines. Load changes are an adverse-event test rather than a load-source output. Their Sections II and IV target short-circuit identification before clearing, whereas our restricted task follows load changes and line openings through restoration and later interventions. Sparse sensing, a physical trajectory library and retrospective onset scanning are shared ideas. We have not run their procedure on the present data and make no numerical superiority claim over it.

Sequential diagnosis also has a stronger statistical literature than a rolling-window label suggests. The Adaptive Matrix CuSum addresses misidentification caused by pre-change observations and establishes delay/error results under its specified observation laws [13]. R11's pruned conditional histories, colored measurement state and plug-in startup estimates do not inherit those guarantees. A ten-second window or the array M is not a replacement for that theory.

Learned PMU diagnosis provides a different comparator. Sparse classifiers and knowledge-driven representations iden-

tify represented events or assets [4], [5], [14]. The historical SGSMA baseline supplies two frozen ExtraTrees adaptations here. We distinguish interpolation at represented sources, transfer of that frozen adapter, and true held-source generalization. The present causal catalog is known; its results do not establish the last of those claims.

Observability and diagnosability explain why neither a physical nor learned method is guaranteed to localize every visible event. Functional observability concerns the requested state function [15], [16]; descriptor-model diagnosability and nuisance-profiled tests concern competing explanations [17], [18]. Placement-dependent source ambiguity is also established in PMU fault/event work [19]–[21]. Our state/source distinction applies these ideas; it is not their first discovery.

Multiple-model filtering already manages conditional state/mode hypotheses [22]; R11 is a pruned history bank, not a new IMM derivation. Stochastic swing/kernel models estimate latent states in Ensaf and Barati [23], and physics-derived prior covariances support sparse-state forecasting [24]. Joint DAE-aware generator/network parameter inference is studied separately [25]. R11 uses a finite bank of constant network hypotheses rather than inferring the continuous parameter posterior. A Gaussian process or neural architecture would require separate validation.

The remaining question is operational and falsifiable: can a state-conditioned bank distinguish a new intervention from a preceding intervention's recovery while retaining timely family/source decisions under fixed network uncertainty? Our contribution is the specified recovery-aware construction and its bounded evidence, not a claim to have invented model-based diagnosis. We use "physics-informed" because physical equations define candidate trajectories and likelihood inputs. Learned discrepancy and the unrestricted joint state/event/integrity estimator remain separate proposals.

III. THE DECISION REQUIRED FROM EIGHT PMUS

At every report, the task is to decide whether the recent trajectory contains a physical intervention and, if so, identify its family, source and onset. Detection is the existence decision; classification chooses the physical operator; localization chooses a bus or branch. Those outputs need not become reliable at the same time. We keep the estimated physical onset $\widehat{\tau}$ separate from the time at which an alarm or diagnosis is emitted.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_q(x_k, z_k; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The measurement map includes gains, phase references, orientation and the causal frequency/ROCOF estimator. A machine's rotor speed is not substituted for bus frequency.

An intervention hypothesis is $h = (e, c, \tau, a, d)$: family, source, onset, severity and duration. A load changes its local P/Q setpoints. A line opening switches its branch primitive; it is not approximated as a load change. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. FROM PHYSICAL PREDICTIONS TO CAUSAL ATTRIBUTION

A. WHERE PHYSICS ENTERS THE ESTIMATOR

The causal bank is the frozen R11 research implementation. Physics determines the candidate response, rather than supplying a penalty to a learned bus-label classifier. Kirchhoff constraints connect each unmeasured source to all available PMUs. With complex bus voltage V, the relevant electrical maps include

$$I_{bus} = Y_{bus} V, \quad I_{term,r} = Y_{term,r} V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal-current rows retain transformer ratios, charging and orientation. A terminal current is not the bus injection in the power equation. A line hypothesis changes both the network constraints and the resulting terminal observations; a load hypothesis changes the local input and its dynamic propagation.

On a regular index-one DAE chart, differentiating the algebraic constraint gives the local predictor

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= (f_x - f_z g_z^{-1} g_x)\delta x + (f_u - f_z g_z^{-1} g_u)\delta u. \end{aligned} \tag{6}$$

TABLE 1. Closest inference contracts and the difference tested here. This is a methodological comparison; only Chen and the frozen ExtraTrees adapters are executed comparators in C01. Absence from a paper's evaluated task is not a claim of algorithmic impossibility.

Route	Physical/statistical mechanism	Published target	Difference or role here
Chen / Rovatsos	Increment statistics, network models and sequential tests	Line outages; Rovatsos includes transient phases	Chen is executed on persistent-line first stops
Warner high-fidelity	Simulated V/I paths, moving onset window, distance evidence	Short-circuit line candidates	Nearest trajectory precedent; no common-data run here
Adaptive Matrix CuSum	Pairwise sequential diagnosis and error/delay analysis	One unknown change among distributional alternatives	No corresponding optimality guarantee is claimed here
DAE / stochastic-state estimation	Incomplete DAE observers or physical uncertainty models	States, unknown inputs or network parameters	Components/adjacent tasks, not episode attribution scores
Knowledge-driven learned diagnosis	Spatiotemporal representation with topology/type constraints	Multi-family disturbance location	Distinct from the physical likelihood; ExtraTrees is our executed learned reference
Physical history bank here	Estimated-past-state paths, active/cause masses, fixed R/X mixture	Successive known-catalog load/line interventions	Test whether recovery-aware attribution retains timely sources without excess alarms

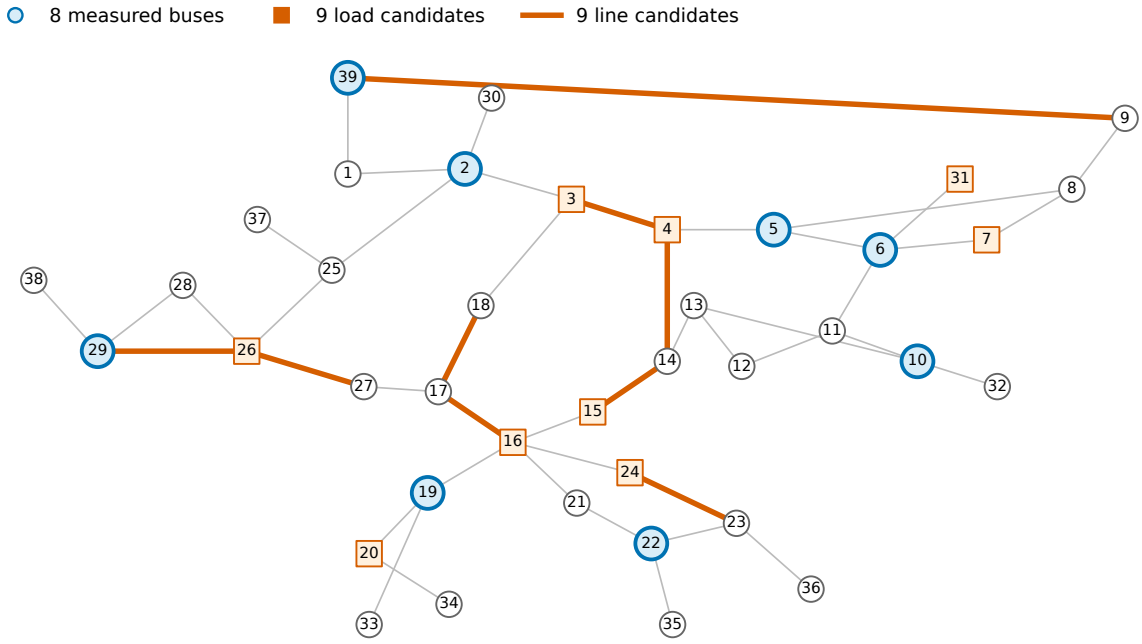


FIGURE 2. What is measured and what is hypothesized. The frozen causal bank contains nine load buses and nine line assets within the 39-bus network; eight blue buses carry PMUs. An unmeasured candidate is inferred through its network response at those PMUs. Orange branches are candidate interventions, not current-sensor assignments. Coordinates are schematic and do not define electrical distance.

The frozen implementation builds transition/observation operators from the full DAE derivatives, with native terminal-current dependence for each topology. Equation (6) explains why an input at an unmeasured bus leaves a network-wide signature. It also explains why a local magnitude threshold alone cannot identify the source. Generator inertia, controls and the load law remain those of the frozen plant; no event-specific plant recalibration occurs here.

B. ASK WHAT COULD HAVE HAPPENED IN THE LAST TEN SECONDS

At report time t, screening considers candidate onsets in $[t - 10\,\mathrm{s}, t]$. For each source and family it asks: if this intervention began at τ, what reports should have arrived from τ to t? Prediction starts from the stored no-event state estimate at τ, not a reset to nominal equilibrium. This matters when a new intervention arrives during recovery from the preceding one.

For a candidate history, the comparison is between a phys-

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside C01
Unknown	Unresolved diagnosis	Not a new physical family

ical prediction and the same observed window,

$$r_{h,W} = y_W - \widehat{y}_{h,W},$$
$$L_{h,W} = \int p(y_W \mid h, a, \mathcal{F}_{\tau^-}) p(a \mid h, \mathcal{F}_{\tau^-})\, da. \tag{7}$$

Conditional state filtering accounts for the retained physical state and colored instrumental noise. Load amplitude is integrated through the inherited truncated-Gaussian calculation; opening a line uses a switched operator. A finite line-screening response proposes a candidate, and the full switched operator confirms it from the estimated initial state. The rolling screen proposes useful histories; it is not additional independent evidence. A selected suffix is replayed with its original state snapshot, while the sequential likelihood scores each report once.

Supplementary Appendix I gives the scalar precision/mean update and normalizing integral used for a retained load history. It makes the calculation reproducible without presenting standard Gaussian square completion as a new inference theorem.

C. KEEP REFERENCE ERROR SEPARATE FROM EVENT DYNAMICS

A wrong observation center can make a quiet network look disturbed. For each fixed network hypothesis θ_ℓ, startup estimates

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{8}$$

Angular differences use the common reference consistently. Frequency and ROCOF already have empirical startup centers and receive no extra intercept. The intercept is then frozen: it cannot follow a later load step and erase its signature. This is an observation-discrepancy correction, not recovery of the true impedances or a substitute for an AC-feasible hidden-state estimate. Its uncertainty and the operating-point estimate are treated by substitution.

The preservation mechanism is elementary but useful. For a fixed center, $T_b(y) = y - y^{ref} - b$ satisfies $T_b(y+s) - T_b(y) = s$: a physical response s is not projected away. Updating that center during an intervention would not retain this guarantee. Freezing the center preserves signatures; it does not repair wrong dynamic sensitivities.

Five constant network hypotheses represent nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X changes. Each propagates its own conditional histories. The outer weight uses a predictive density formed before the current-report state update or screening:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{9}$$

Startup reports are not scored again for network-model selection. Pruning, shared covariance approximations and adaptive history proposals make the result an approximate prequential mixture. We do not claim an optimal Kalman filter, a calibrated event posterior or identification of all line parameters.

D. WHAT M AND ´M ACTUALLY DO

Let c index an admissible family/source cell. For a bus load the physical index is diagonal, (i, i, load); for a line it is (i, j, line). The implementation stores only the 18 allowed cells. It does not infer every event at every one of the 39 buses.

For network model ℓ, let $M_{\ell,c,k}$ be the active history mass assigned to cell c. The mixed array and its exact increment are

$$M_{c,k} = \sum_\ell w_{\ell,k} M_{\ell,c,k},$$
$$\Delta M_{c,k} = \underbrace{\sum_\ell w_{\ell,k} (M_{\ell,c,k} - M_{\ell,c,k-1})}_{\text{conditional evidence evolution}} + \underbrace{\sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}}_{\text{network-model reweighting}}. \tag{10}$$

The array is operational: the mixed masses drive decisions. The decomposition identifies whether an attribution changed within the conditional observers or because network hypotheses were reweighted. It is an accounting identity checked by the code, not a new tensor-factorization theorem. A separate cause mass retains the originating intervention after it switches OFF, so the observer can explain recovery without declaring a new event.

The nominal reference runs the same physical/measurement construction with one network hypothesis and its own publication path. Comparing it with the five-model version tests the value and cost of model uncertainty without changing the plant or its source catalog.

E. WHY RESTORATION DOES NOT REMOVE THE PHYSICAL EXPLANATION

Let κ be a candidate restoration time. For the supported load and line operators, the differential state is continuous even though the algebraic solution can jump: $x(\kappa^+) = x(\kappa^-)$. Restoring the input or topology therefore does not restore the pre-event state. On a regular, locally linear post-restoration

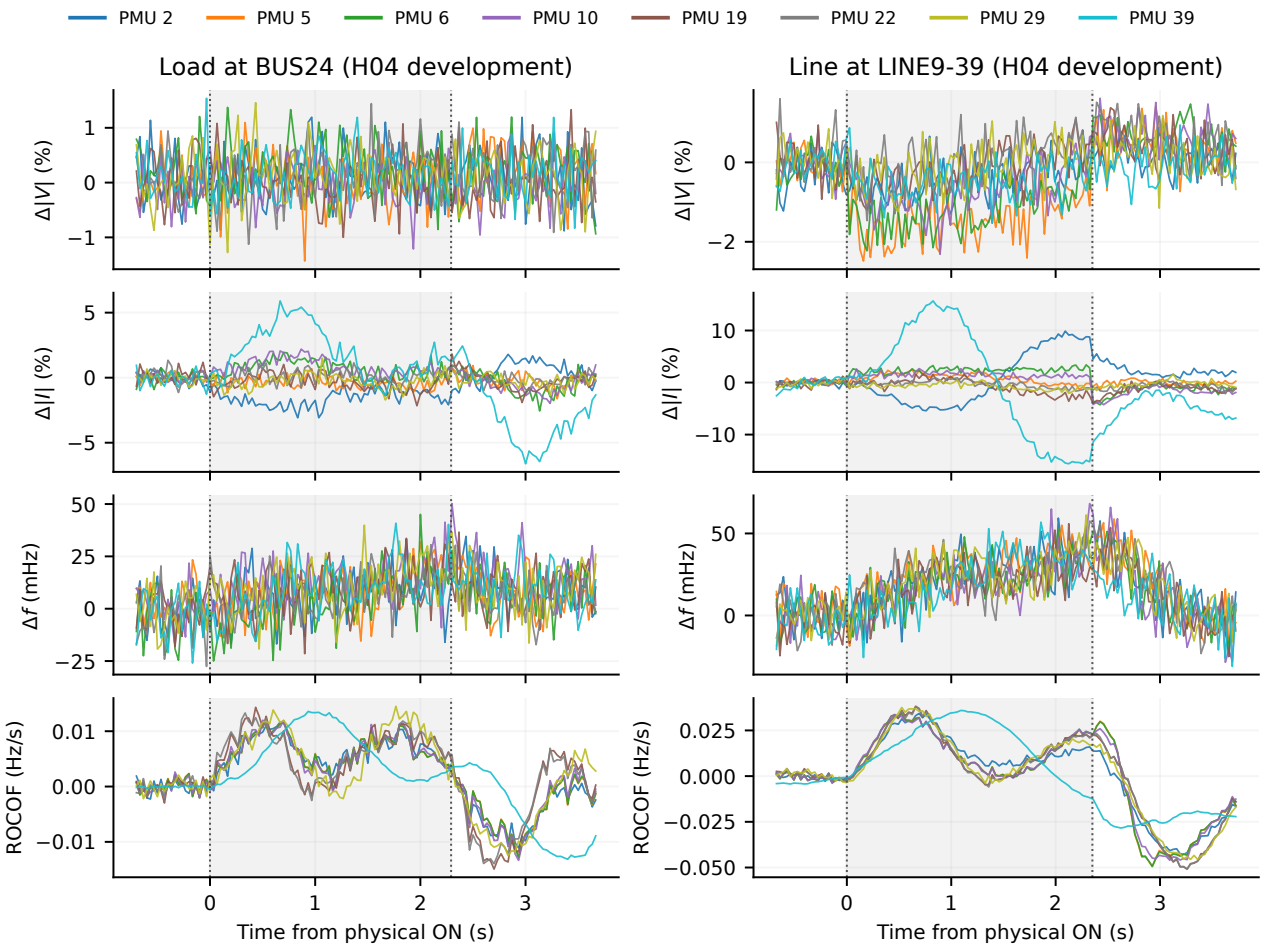


FIGURE 3. **Why the method compares network responses. Two fixed H04 development examples show the first load and first line intervention in parent H002, measured at all eight PMUs. Shading marks the true intervention interval for retrospective explanation only. The different V/I/frequency/ROCOF responses are observations, not universal family templates: source, operating point and severity also change their shape. Frequency and ROCOF include the frozen causal instrumental estimator.**

chart, compare the restored candidate with the same-context no-new-intervention trajectory. Their state/output differences satisfy

$$\delta x_h(t) \simeq \Phi_0(t,\kappa)\delta x_h(\kappa),$$
$$\delta q_h(t) \simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa, \tag{11}$$

where q denotes the network voltage/terminal-current coordinates before the frozen instrumental map. Φ_0 propagates the restored DAE's reduced dynamics and C_0 is its reduced observation map. The actual likelihood also retains the measurement/filter memory. Thus the baseline during recovery is a predicted trajectory, not a constant or zero residual.

Equation (11) is a local interpretation of retained state propagation, not a new propagation law or an additional fitted model. R11 keeps active-event mass separately from originating-cause mass. A new intervention must compete with the existing recovery explanation; its onset does not justify resetting either machine or instrumental states. The H04 time trace illustrates this construction. The nominal comparison isolates the outer network bank, while the admission ablation changes a publication decision; neither alone isolates the causal benefit of every recovery-state operation.

F. EMIT DECISIONS WITHOUT RESETTING THE NETWORK

An alarm, a confirmed family/source diagnosis and an estimated restoration are separate messages. The inherited configuration uses alarm/restoration mass thresholds of 0.99, six confirmation reports for alarm evidence, and 15 reports for restoration confirmation. These numbers are frozen method settings, not calibration guarantees. Additional source/line confirmation and episode-replacement rules remain those of the archived implementation; the configuration is included in the evidence snapshot.

Every emitted message records the latest report used. The observer continues through OFF with its conditional state and covariance; it does not reset to an equilibrium. A later event can begin while the network is still recovering. A source diagnosis emitted after true OFF remains correct eventually but fails the timely-localization metric.

The restricted runtime algorithm is: initialize from the

permitted prefix; predict each fixed-model history; score the new report; propose/replay retrospective interventions when supported; mix the history masses; publish a confirmed decision; retain states through restoration. This is model-based Bayesian hypothesis inference. The ExtraTrees comparators are the learned components in this comparison; R11 has no learned bus-label head.

V. WHEN A PHYSICAL EXPLANATION CAN BE DISTINGUISHED

A physical model does not remove an information limit. Two sources can produce nearly the same reports after their amplitudes and initial conditions are allowed to vary. High confidence in the state estimate therefore need not imply high confidence in a source.

The distinction is explicit in a local model. With observation matrix C, transition A and requested hidden-voltage map L, functional reconstruction requires

$$\ker \mathcal{O}_H \subseteq \ker L, \qquad \mathcal{O}_H = \operatorname{col}(C, CA, \ldots, CA^H). \quad (12)$$

Unobservable directions are harmless for this target if they do not change Lx [15], [16]. Source attribution asks a stronger and different question. Let $s_i(\eta_i)$ and $s_j(\eta_j)$ be candidate trajectories after permitting their nuisance coordinates η, including severity and the admitted initial-state variation. Their profiled separation is

$$D_{ij}(H) = \frac{1}{2} \min_{\eta_i, \eta_j} \|s_i(\eta_i) - s_j(\eta_j)\|^2_{R_H^{-1}}. \quad (13)$$

Under a common Gaussian observation law, an intersection gives identical measurement distributions at that intersection. No classifier can then distinguish the candidates uniformly over the allowed nuisance set. A small positive distance can still leave a difficult, noise-sensitive decision [17], [18]. The detailed finite-horizon, signed-support and physical-boundary qualifications remain in Supplementary Appendix D.

These are local conditions, not a structural certificate for the tested IEEE-39 configuration. The archived thresholded Gramian audit has a nonmonotone pseudo-rank and cannot supply that certificate. The nonlinear state-recovery result is evaluated independently; accurate target reconstruction is not used to claim full-state observability.

This framework explains the role of longer history: it may separate response paths that are close in one snapshot, but cannot guarantee an easy decision for every source and severity. It also explains why the estimator must keep competing hypotheses rather than turn every detected change into a confident location.

We test this mechanism with the separate batch reference $\mathcal{E}_{\mathrm{B}}$. It receives 32 rectangular voltage/current coordinates and is given the onset, load family and operating point; it compares 137 zero-, one- and two-source load supports and retains the original GH31 amplitude integration and W2 covariance. It is not R11 with a different solver, and R11 is not silently declared the complete joint estimator $\mathcal{E}_{\mathrm{J}}$. The batch study isolates source ambiguity; the causal study tests sequential decisions under a different contract.

VI. EVIDENCE DESIGN AND FAIR COMPARISONS

The experiments answer three successive questions: does the physical model recover useful electrical information; are source responses distinguishable after nuisance fitting; and does a causal decision retain precision when interventions and recovery follow one another? They are separate tests, not parts of one pooled accuracy score.

The causal simulator uses the frozen IEEE-39 Julia/PowerDynamics candidate, with continuous states and a separate PMU reporting grid. A 120-s record contains 3601 reports per PMU; a 600-s normal record contains 18001. Event-aligned reports use the consistent post-switch algebraic value, and solver diagnostics are stored separately. The measurement references, load law and stochastic instrument parameters are not retuned to individual events. RAW0001 is retrospective calibration evidence; RAW0002 remains unopened.

H03 and H04 each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

R11 and its nominal version are frozen co-references. Both ExtraTrees artifacts and their causal preprocessing/publication adapters are frozen as well; neither is retrained on H04 or C01. They receive the same 30-fps observations. This is a comparison of those operational implementations on the stated contract, not a rerun of the hackathon leaderboard.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [6]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared F01 adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all F01 instrumental effects. We preserve both matched-law implementation checks and F01 mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. It is not assigned a load classifier or a restoration operator. C01 therefore has two evaluation lanes: continuous load/line episodes for R11/nominal/ML, and first-stop persistent-line decisions for all comparators. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with

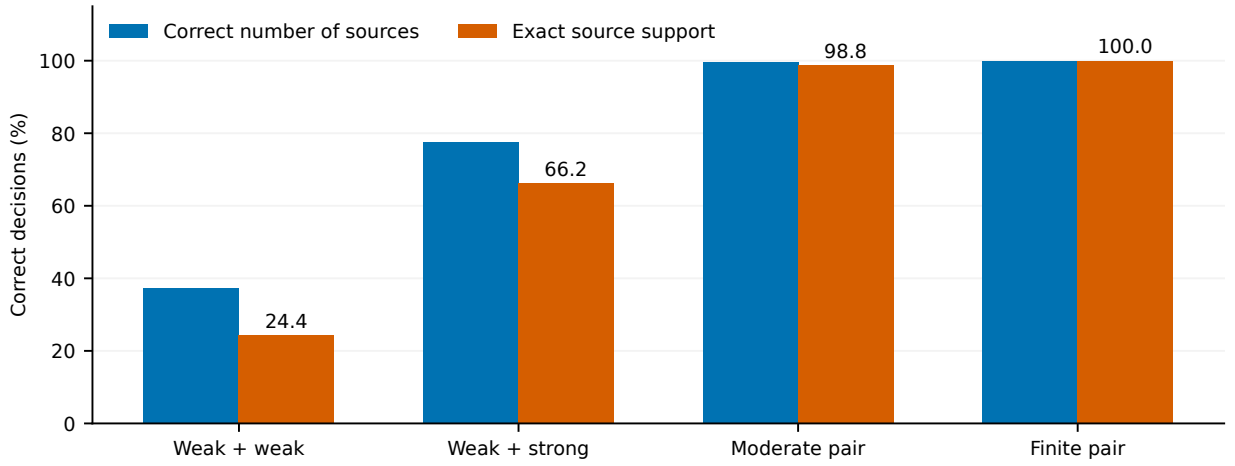


FIGURE 4. **The resolution limit behind the attribution problem. The frozen 137-hypothesis load bank increasingly recovers both cardinality and exact support as event strength increases. Weak-pair exact-support Top-1 is much lower than finite-pair Top-1. Cardinality means choosing the correct number of sources; support means choosing their identities. These known-onset batch results explain ambiguity and do not certify R11's causal performance. Repeated noise decisions are grouped by physical trajectory in the archived protocol.**

TABLE 3. **One argument, distinct evidence contracts. An event realization repeats a physical parent under a condition or noise draw; it is not another independent physical event.**

Question	Executed experiment	Decisive outcome	What it does not establish
Useful state information?	Separate B0–B3 voltage reconstruction and AC recentering	Hidden-voltage error and model-shift sensitivity	Source identity or calibrated joint covariance
Which sources are distinguishable?	Prospective 137-support batch load bank; known onset	Cardinality and exact-support recovery by severity	Unknown-onset multi-family monitoring
Can it monitor successive events?	H03/H04 continuous load/line development	Timely family/source, extras and restoration behavior	Confirmatory superiority or unseen-source transfer
Does the frozen result persist?	C01 configuration sealed before its new predictions	Independent-parent, paired R/X/noise comparison	No result is claimed until execution completes

the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

C01 specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three fixed R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories produce 342 intended observed records. All solver and inference failures remain in the ledger. Comparisons use 10,000 paired parent-cluster resamples; conditions and noise realizations do not create extra independent parents. Predictions are sealed before the evaluator opens reserved truth. Until that evaluation exists, the following section reports only completed development evidence.

VII. WHAT WORKS, WHAT FAILS, AND WHY

A. PHYSICAL STATE RECOVERY IS USEFUL BUT DOES NOT LOCATE AN EVENT

The separate matched-model state study reconstructs 31 unmeasured bus voltages with 0.009634% mean relative error. Under operating-point shift, regularized AC-feasible recentering closes 97.4%–98.3% of the tested oracle gap. Yet the nuisance information has rank 25 of 43, and its joint covariance diagnostic remains unreliable. The useful conclusion is target recovery under a stated model, not exact parameter identification or a source-localization guarantee. Full state/mismatch tables and diagnostics remain in the supplement and frozen evidence.

B. PHYSICAL SIGNATURES RESOLVE STRONGER SOURCES, NOT EVERY WEAK PAIR

The batch reference evaluates 137 hypotheses on 3488 fresh nonlinear trajectories. Exact-support Top-1 rises from 24.4% for weak pairs to 100.0% for finite pairs. In the weak regime, even cardinality is imperfect; it nevertheless resolves more often than exact source identity (Fig. 4). This is the distinction in (13): a visible intervention need not have a well-separated location.

The archived retrospective audit associates errors with the closest competing response after amplitude fitting, rather than with a support-only geometric score. The longer-horizon diagnostic increases separation but leaves strongly collinear candidates. Those studies explain a boundary; they do not turn a local numerical condition into a global observability certifi-

TABLE 4. What the causal component actually resolves in H04 development. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

cate. The original GH31 batch estimator and its denominators remain unchanged.

C. THE CAUSAL GAIN IS IN LOAD RECOGNITION UNDER THIS CONTRACT

The family breakdown in Table 4 is more informative than an aggregate score. In H04, both frozen ExtraTrees adapters and both physical references identify all tested line interventions in time. The physical bank's advantage is concentrated in loads: the five-model version retains 26/27 timely load decisions and its nominal version 27/27, whereas these particular ExtraTrees adapters retain 0/27. This is an observed transfer failure of those frozen adapters on this contract, not proof that supervised learning cannot detect loads. It also cannot be substituted for their historical multi-class hackathon score.

The five-model physical reference detects and eventually attributes 54/54 realizations, but only 53 diagnoses arrive before physical OFF. Its nominal reference has 54 timely diagnoses. Thus detection, eventual attribution and timely attribution give different answers on the same stream. The nominal reference's additional timely decision comes with more extras: 7 versus 4. Model uncertainty has a measured trade-off, not a uniform benefit.

D. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 6 exposes the sequence hidden by a count. A load intervention changes several PMU channels, the history bank accumulates evidence, and the active source mass changes. After OFF, the physical state continues evolving. The retained cause/recovery histories allow the observer to explain that trajectory without resetting the network. A later source must compete against both the no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and (10) separates that revision from within-model evidence. This is what the array adds to the explanation. The figure does not establish that a tensor representation alone improves accuracy.

An additional onset-witness publication ablation reaches zero extras while reducing timely attribution to 52/54. It rejects a true BUS16 load with weak early onset evidence. The physical states and underlying masses are inherited, so suppressing the public alarm does not repair the estimator's explanation. We retain this as a negative experiment rather than select it for its zero-extra count. R11 and nominal remain frozen for C01.

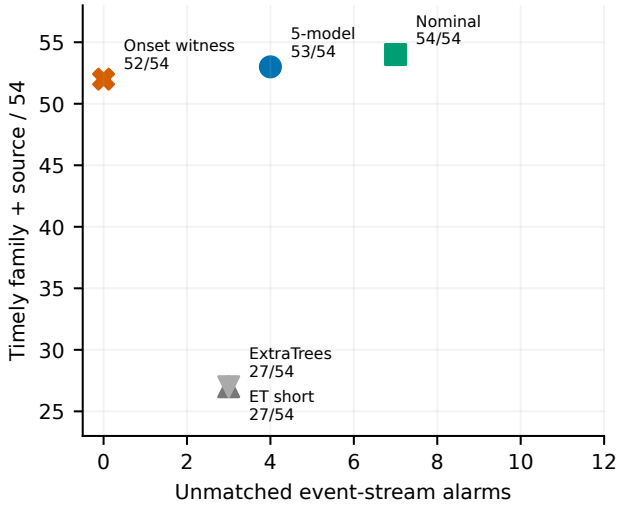


FIGURE 5. The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

E. WHAT IS STILL BEING CONFIRMED

The present development evidence supports causal load/line attribution on a known catalog and shows its failure modes. It does not yet support a final performance claim on C01. The confirmatory executor is running with fixed methods, cases, thresholds and metrics; its completed results will enter the dedicated section below through the same hash-to-table/figure chain. No partial C01 result is used here to choose a variant or revise the method.

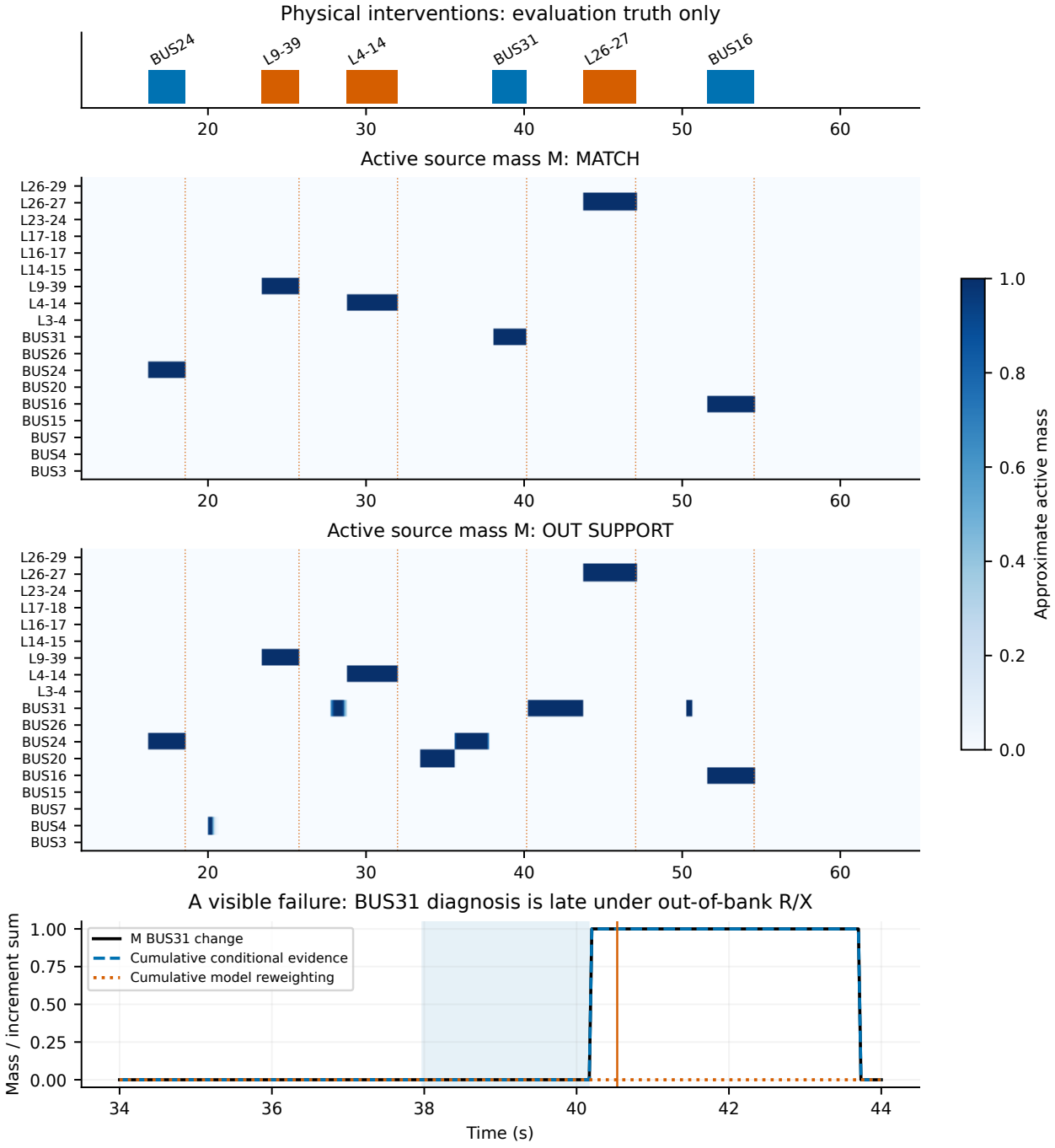


FIGURE 6. **From a sequence to an attribution array. Top: reserved physical intervention intervals in fixed H04 parent H002. Middle: causal active masses across the 18 admissible cells for paired nominal and out-of-bank records. Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 under out-of-bank R/X, with the exact cumulative decomposition of $\acute{M}$ relative to 34 s into conditional evidence and model reweighting. The diagnosis marker is later than OFF: an eventual correct source is not a timely success. This is a development illustration, not C01 confirmation or a calibrated probability display.**

VIII. INTERPRETATION AND LIMITS

The bank constructs each candidate from the network equations rather than fitting a source-label decision boundary. That is an architectural distinction, not evidence of causal source-excluded generalization. It retains the explanation as states evolve and distinguishes an operator being restored from the grid returning to equilibrium. In this development test the practical gain is timely load attribution; line attribution does not separate the physical and learned references.

That conclusion has three boundaries. First, the causal result uses a known nine-load/nine-line catalog and an intervention-free startup. It is not a test of all 39 buses or arbitrary unknown event families. Second, the instrument and terminal map are frozen research assumptions, including the provisional BUS5 assignment and excluded BUS39 current angle. Agreement within that model does not establish the producer's actual CT placement or exact replication of RAW0001. Third, the network bank is finite and its startup nuisance estimates are plug-in. Its masses are useful attribution weights but are not demonstrated calibrated probabilities.

The nominal reference and failed onset-witness ablation prevent a stronger accuracy claim. More network hypotheses reduce extras in H04 but delay a BUS31 decision; more restrictive admission removes extras but rejects a real BUS16 event. Those are differences in what the decision retains and when it is emitted. Reporting eventual source recovery, or zero extras alone, would select different winners and hide the trade-off.

The mathematical interpretation is equally scoped. Good hidden-voltage reconstruction does not prove event separability. The batch test supports the importance of profiled response overlap, but does not validate every analytic small-amplitude or changing-sensor limit in the supplement. The 137-support batch reference retains its original quadrature; the causal implementation has not been proved to be the unrestricted joint estimator or an exact reduction of it.

The ExtraTrees result tests the particular frozen causal adapters under these physical conditions. Their load failure calls for that restricted interpretation, rather than a general statement that ML cannot recognize loads. Likewise, Chen's colored-noise/DC mismatch is kept visible instead of being used to claim universal superiority over the published algorithm under its own assumptions.

The closest-method comparison narrows the novelty claim further. Warner's trajectory scan already supplies physical transient signatures, unknown-onset search and sparse remote sensing; the Adaptive Matrix CuSum supplies sequential diagnosis theory. Our evaluated construction addresses a different operational target: successive load/line histories whose restoration and electrical recovery are distinct. The present evidence does not establish superiority over those unexecuted comparators or a false-alarm guarantee. Nor does the component-level source geometry yet provide a calibrated online abstention certificate for R11.

A physics-informed extension should preserve this distinction rather than merely add a learned label head. Physics-derived stochastic priors can propagate uncertainty in unmeasured forcing [23], [24], while joint DAE parameter inference addresses a different identification problem [25]. A future discrepancy model would have to separate such uncertainty from intervention signatures and instrumental noise. That is a new validation question, not a feature silently added to the frozen C01 method or a condition introduced to postpone this paper's closure.

The final decision rests on the prespecified C01 comparison and claim/evidence review, without another bus-specific correction. Faults, generation mechanisms, PMU loss, unseen-source causal transfer and field operation are future scopes, not hidden requirements for reporting this restricted result. The proposed joint estimator remains documented in the supplement so that exploratory work is preserved without being mistaken for the mechanism validated here.

IX. CONCLUSION

The contribution is a recovery-aware physical attribution construction for sparse PMUs: a restored operator leaves a dynamic state that continues to explain measurements and must compete with later interventions. The implemented load/line bank predicts intervention-conditioned DAE histories and mixes fixed network hypotheses. M records the source masses; ΔM separates within-model evolution from model reweighting. Physical templates and unknown-onset scans have close precedents, and this study does not establish superiority over unexecuted trajectory-based methods.

The completed development evidence shows both utility and a limit. Physical state recovery can be accurate while weak event sources remain ambiguous. In causal H04 development, the five-model reference obtains 53/54 timely joint decisions with 4 extras; nominal improves timeliness at a higher alarm burden. The family and temporal breakdowns explain that result, including the BUS31 delay and the negative BUS16 admission experiment. Final performance claims depend on the frozen confirmatory results. No broader all-family, unseen-source or field claim follows from these tests.

SUPPLEMENTARY MATERIAL

The supplement preserves detailed proofs, electrical and instrumental conventions, the original batch estimator and joint-estimator proposals, expanded component studies, and complete evidence contracts. The pre-revision manuscript is archived in the repository. Empirical data and inference baselines are unchanged.

REFERENCES

[1] A. G. Phadke and J. S. Thorp, *Synchronized Phasor Measurements and Their Applications*. New York, NY, USA: Springer, 2008.

[2] J. Zhao, J. Qi, Z. Huang, A. P. S. Meliopoulos, A. Gómez-Expósito, M. Netto *et al.*, "Power system dynamic state estimation: Motivations, definitions, methodologies and future work," *IEEE Transactions on Power Systems*, vol. 34, no. 4, pp. 3188–3198, 2019.

[3] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.

[4] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
[5] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
[6] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.
[7] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.
[8] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023, posted June 16, 2023; cited as a preprint.
[9] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.
[10] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.
[11] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.
[12] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.
[13] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.
[14] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
[15] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.
[16] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.
[17] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.
[18] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.
[19] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.
[20] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.
[21] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.
[22] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.
[23] M. Ensaf and M. Barati, "Harnessing kernel regression for stochastic state estimation in solar-integrated power grids," arXiv preprint arXiv:2309.10200, 2023, version 1. [Online]. Available: https://arxiv.org/abs/2309.10200
[24] A. M. Tartakovsky, T. Ma, D. A. Barajas-Solano, and R. Tipireddy, "Physics-informed gaussian process regression for states estimation and forecasting in power grids," *International Journal of Forecasting*, vol. 39, no. 2, pp. 967–980, 2023.
[25] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," arXiv preprint arXiv:2604.15686, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2604.15686

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Supplementary Material for Recovery-Aware Physical Event Attribution From Sparse PMUs

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ABSTRACT This supplement records the detailed literature matrix, electrical conventions, mathematical proofs, Fisher-information analysis, Bayesian estimator construction, frozen experimental contracts, and secondary results supporting the main article. It distinguishes the independently evaluated state component and implemented batch event estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. The latter remains unexecuted. This supplement does not add empirical claims or change any test denominator. Every empirical table and figure is generated from the same hashed evidence snapshot used by the main manuscript; separate deterministic tables audit stated algebraic identities.

INDEX TERMS Bayesian inference, Fisher information, functional observability, phasor measurement units, reproducibility, state estimation, event-source inference.

APPENDIX A DETAILED LITERATURE CAPABILITY MATRIX

Table 1 expands the family-level synthesis in the main paper. "Held source" means that the physical asset is absent from source-labeled training and tuning. The entries describe published tasks and assumptions; they are not a cross-paper performance ranking.

The direct competitors solve different contracts. State and unknown-input estimators define the latent reconstruction problem but not the event taxonomy. Learned localizers rank represented sources but rarely remove the physical asset from every source-labeled fitting operation. Model-based outage methods provide the closest likelihood precedent, while functional observability and placement studies provide the information tests. These distinctions determine the baselines and prevent cross-paper accuracy comparisons with incompatible targets.

APPENDIX B ELECTRICAL MODEL AND EVENT OPERATORS

Within a smooth network mode, the DAE is locally index one after fixing the reference angle. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1}(g_x f + g_u \dot{u} + g_{\boldsymbol{\theta}} \dot{\boldsymbol{\theta}}). \tag{1}$$

This reduction is not applied across a topology switch or fault insertion, where algebraic variables may jump. Differential states remain continuous unless a simulator-specific reset is declared.

With $Y_{\mathrm{bus}} = G + \mathrm{j}B$ and $V_i = |V_i| e^{\mathrm{j}\vartheta_i}$, the network injections are

$$P_i = |V_i| \sum_j |V_j| (G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}), \tag{2}$$

$$Q_i = |V_i| \sum_j |V_j| (G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}). \tag{3}$$

For branch $e = (i,j)$ with complex tap t_e, series admittance y_e, and terminal shunts $y_e^{f,sh}$ and $y_e^{t,sh}$,

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \tag{4}$$

The implemented PMU operator selects the actual terminal and orientation. Replacing this current with complete bus-injection current changes the likelihood.

Positive-sequence phasors define the balanced model. Negative- and zero-sequence components remain integrity or unbalance evidence. Ideal continuous-frequency channels satisfy

$$f_i^{\mathrm{phys}} = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \qquad \mathrm{ROCOF}_i^{\mathrm{phys}} = \frac{1}{2\pi} \frac{d^2 \angle V_i}{dt^2}, \tag{5}$$

but an operational likelihood must reproduce the causal filtering of the PMU backend.

TABLE 1. Detailed capability comparison for the closest methodological works.

Work	State target	Event/source target	Physics mechanism	Sparse sensing	Held source	Integrity treatment
Katanic *et al.* [1]	Differential and algebraic DAE states	Unknown injectors	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Warner *et al.* [2]	Simulated transient paths	Short-circuit line candidates	DAE V/I trajectories and moving onset scan	Two PMUs in IEEE-39 case	Not a source-exclusion test	Load forecast/adverse-load tests
Chen / Rovatsos [3], [4]	Network angle increments	Outage identity	QCD; Rovatsos includes transient phases	PMU subset	Not a learned source task	Stochastic injection law
Warner / Fellouris [5]	General sequential observations	Post-change alternative	Adaptive Matrix CuSum	Not a PMU placement study	Finite specified alternatives	Error/delay theory under stated laws
Nugroho *et al.* [6]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance
Qu *et al.* [7]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo *et al.* [8]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue *et al.* [9]	Distribution-system state	Line outage	Grid-model-informed graph learning	Yes	Not stated	Outlier-resistant SE
Moshtagh *et al.* [10]	Transmission-system state	Topology variation	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad data
Pandey *et al.* [11]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [12]	None	Fault detection and location	OLS/CNN and PMU placement	Yes	Not stated	No
Li *et al.* [13]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow *et al.* [14]	None	Known/unknown event type and location	Siamese distance and learned rejection	Limited/full sets	Open-set rejection	Rejection boundary
Hu and Cheng [15]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No source posterior
Rimkus *et al.* [16]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	Imputer only	Corruption via imputation
Zhai *et al.* [17]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmarks	Attack target only
R11 in main article	Retained conditional states	Successive nine-load/nine-line catalog	DAE histories, recovery and fixed R/X mixture	Eight PMU buses	Not tested; no source-label head	Colored instrument; plug-in startup
Separate batch component	Known-context voltage/current	Load supports with $K \leq 2$	Original GH31/W2 likelihood	Eight PMU buses	Different contract from R11	Matched frozen noise

The preceding equations define the ideal physical quantities f_i^{phys} and $\mathrm{ROCOF}_i^{\mathrm{phys}}$ implied by the network trajectory. Reported quantities f_i^{meas} and $\mathrm{ROCOF}_i^{\mathrm{meas}}$, however, are outputs of a dynamic frequency-estimation process and can exhibit disturbance- and estimator-dependent transient error and latency. A prior benchmark by the authors compared 16 dynamic frequency estimators spanning six practical groups over 34 disturbance scenarios and quantified these trade-offs under ramps, phase jumps, harmonics, ringdown, and composite IBR disturbances [18]. We therefore distinguish

$$\begin{bmatrix} f_i^{\mathrm{meas}} \\ \mathrm{ROCOF}_i^{\mathrm{meas}} \end{bmatrix} = \begin{bmatrix} f_i^{\mathrm{phys}} \\ \mathrm{ROCOF}_i^{\mathrm{phys}} \end{bmatrix} + \boldsymbol{\varepsilon}_{fr,i}, \qquad \mathrm{Cov}(\boldsymbol{\varepsilon}_{fr,i}) = \boldsymbol{\Omega}_{fr}. \tag{6}$$

Because phasor, frequency, and ROCOF outputs may be estimated from overlapping underlying waveform or phasor information, their errors need not be conditionally independent. An augmented likelihood therefore permits cross-covariance

$$\boldsymbol{\Omega}_{\mathrm{aug}} = \begin{bmatrix} \boldsymbol{\Omega}_{VI} & \boldsymbol{\Omega}_{VI,fr} \\ \boldsymbol{\Omega}_{fr,VI} & \boldsymbol{\Omega}_{fr} \end{bmatrix}; \tag{7}$$

neither the magnitude of $\boldsymbol{\Omega}_{VI,fr}$ nor a block-diagonal approximation is assumed without calibration. Frequency and ROCOF are retained as candidate channels for an augmented sparse-PMU likelihood. Their incremental source-discrimination value, covariance structure, and effect on localization delay are not assumed here and require the dedicated *VI*, *VI* $+ f$, *VI* $+$ ROCOF, and *VI* $+ f +$ ROCOF ablation.

APPENDIX C METRICS, CAUSALITY, AND EVALUATION UNITS

For trajectory r, hidden-voltage relative error is

$$\mathrm{HVRE}^{(r)} = \frac{1}{T_r|\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\widehat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \tag{8}$$

It is a trajectory-and-bus mean reported as a percentage; it is not the standard per-frame synchrophasor TVE conformity statistic. Detection, event-family macro-F1, exact physical Top-1/Top-3, cardinality, candidate-set coverage, and integrity-source Top-1 retain their own populations. A joint

TABLE 2. Physical intervention operators represented by the common DAE formulation.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive for the clearing interval	admissible buses	complex admittance or resistance/reactance	x continuous; z may jump
Line outage	Remove branch tap, series, charging, and shunt primitives	in-service branches	status and switching time	x continuous; z may jump
Generation change	Modify scheduled injection or a declared governor reference	generator buses	ΔP_G, ΔQ_G, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model	load buses	ΔP_L, ΔQ_L, ZIP convention	continuous input-driven response

outcome is correct only when every applicable component is correct.

A causal estimate at k uses $y_{0:k}$. A fixed-lag smoother, complete-window score, or centered temporal feature is retrospective. For source generalization, the physical asset is absent from source-labeled training, feature fitting, threshold selection, and calibration. Intervals and significance tests use the complete trajectory, event parent, or rebuilt plant as the independent unit; samples within a 30-Hz stream are correlated observations, not replicates.

APPENDIX D PROOFS AND ADDITIONAL DIAGNOSABILITY RESULTS

This appendix develops the mathematical chain used by the paper. It begins with the sampled hybrid event map, separates state-functional recovery from event-source separation, derives the relevant Fisher information after physical nuisance is removed, and then states the finite-horizon and temporal limits. The results are local unless a statement explicitly says otherwise.

A. REGULARITY AND STATISTICAL CONTRACT

Let $u = \mathrm{col}(x, z)$ collect differential and algebraic coordinates and consider the descriptor model

$$E\dot{u} = F(u, p), \qquad y = h(u, p). \tag{9}$$

The analysis uses four assumptions.

1) The reference angle has been fixed and the algebraic Jacobian is nonsingular on the local mode under study, so a regular index-one chart exists between discrete events.
2) The scheduled event time is fixed. In a neighborhood of zero amplitude, the complete callback–solver–sampling map is twice continuously differentiable.
3) Every covariance used as a metric is positive definite on the retained measurement subspace. Candidate-dependent covariances retain their log determinants in likelihood comparisons.
4) The support set is finite and all Bayesian priors are proper. Exact signed-amplitude equivalences use a neighborhood of zero within each support subspace; constrained physical amplitudes instead use their actual difference set.

These assumptions distinguish the local theory from general descriptor left invertibility, state-triggered saltation, and global nonlinear identifiability.

B. SIMULATOR-CONSISTENT EVENT SENSITIVITIES

For a fixed-time, impulse-free event, the differential state is continuous, but the algebraic state must satisfy the post-event constraint. This physical right limit must be distinguished from the values stored inside a simulator callback and from the first sampled frame. The following result makes that distinction explicit.

Theorem 1 (Local consistency-jump expansion): Let $x_0 = x(\tau^-)$ and suppose $g_+(x_0, z_0, 0) = \mathbf{0}$. If g_+ is three times continuously differentiable and g_z is nonsingular at $(x_0, z_0, 0)$, there is a unique local consistent branch

$$z_+(a) = z_0 + az_1 + \tfrac{1}{2}a^2 z_2 + O(a^3), \tag{10}$$

where all derivatives below are evaluated at the nominal right-limit mode and

$$z_1 = -g_z^{-1} g_a, \tag{11}$$

$$z_2 = -g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right). \tag{12}$$

The corresponding differential state satisfies $x(\tau^+) = x(\tau^-)$, and its first event-amplitude sensitivity starts with

$$\dot{x}_a(\tau^+) = f_a - f_z g_z^{-1} g_a \equiv B_{r,a}. \tag{13}$$

Its second-order initial forcing is

$$\begin{aligned}\dot{x}_{aa}(\tau^+) \equiv B_{r,aa} = {} & f_{zz}[z_1, z_1] + 2f_{za}z_1 + f_{aa} \\ & - f_z g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right).\end{aligned} \tag{14}$$

a: Proof.

The implicit-function theorem applied to $g_+(x_0, z_+(a), a) = \mathbf{0}$ gives a unique smooth local branch. Differentiating once gives $g_z z_1 + g_a = \mathbf{0}$, which yields (11). Differentiating a second time gives $g_z z_2 + g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa} = \mathbf{0}$, which yields (12). An impulse-free fixed-time switch does not reset the energy-storage coordinates, so the differential state is continuous. Differentiating $\dot{x} = f_+(x, z_+(a), a)$ at the right limit, with $x_a(\tau^+) = \mathbf{0}$, gives (13). A second differentiation, using $x_{aa}(\tau^+) = \mathbf{0}$ and (12), gives (14). □

The theorem also defines an implementation-independent consistency diagnostic. If a callback changes the parameter but temporarily retains z_0, its algebraic residual is

$$r_{\mathrm{cb}}(a) = g_+(x_0, z_0, a) = ag_a + \tfrac{1}{2}a^2 g_{aa} + O(a^3). \tag{15}$$

Substituting the zeroth-, first-, and second-order approximations to (10) produces constraint residuals of orders $O(a)$,

$O(a^2)$, and $O(a^3)$, respectively. Central finite differences of an independently solved algebraic branch can therefore test both derivatives and the expected convergence orders without relying on undocumented callback storage.

For a differentiable PMU map $y = h(x, z, a)$, the direct right-limit signature is

$$y_a(\tau^+) = h_a - h_z g_z^{-1} g_a, \qquad \Delta y_{\mathcal{P}} = aM_{\mathcal{P}} y_a(\tau^+) + O(a^2). \tag{16}$$

This direct algebraic visibility is not the entire event signature. Even when (16) is nearly zero at the retained PMUs, (13) can generate a distinguishable dynamic response over later frames. Instantaneous invisibility therefore does not imply finite-horizon nonidentifiability.

The frozen artifacts begin at the first retained 30-Hz frame and do not record the callback microstate or an independently solved algebraic right limit. Accordingly, the paper treats the right-limit equations above as the physical contract, not as an observed fact about the current ANDES callback implementation. The relevant differentiable object for the existing response dictionary is the sampled map

$$\mathbf{\Psi}_{T_s}^{\mathrm{sim}} : (u^-, p(a)) \mapsto u(\tau + T_s; a), \qquad T_s = 1/30 \text{ s}. \tag{17}$$

For event coordinates a_i, define

$$L_i = \left.\frac{\partial \mathbf{\Psi}_{T_s}^{\mathrm{sim}}}{\partial a_i}\right|_{a=\mathbf{0}}, \qquad L_{ij}^{(2)} = \left.\frac{\partial^2 \mathbf{\Psi}_{T_s}^{\mathrm{sim}}}{\partial a_i \partial a_j}\right|_{a=\mathbf{0}}. \tag{18}$$

For $t \geq \tau + T_s$, the first and second sensitivities obey

$$E\dot{s}_i = F_u s_i + F_p p_i, \tag{19}$$

$$E\dot{s}_{ij} = F_u s_{ij} + r_{ij}, \tag{20}$$

where

$$r_{ij} = F_{uu}[s_i, s_j] + F_{up}[s_i, p_j] + F_{pu}[p_i, s_j] + F_{pp}[p_i, p_j] + F_p p_{ij}. \tag{21}$$

The initial conditions are $s_i(\tau + T_s) = L_i$ and $s_{ij}(\tau + T_s) = L_{ij}^{(2)}$. For the affine event parameterization, $p_{ij} = \mathbf{0}$.

Theorem 2 (Propagation of event-map mismatch): Suppose two second-order models have the same nominal trajectory, first-order sensitivities, parameter derivatives, and forcing in (20), but different second-order event-map initial conditions. Their difference $e_{ij} = s_{ij}^A - s_{ij}^B$ satisfies

$$E\dot{e}_{ij} = F_u e_{ij}, \qquad e_{ij}(t) = \mathbf{\Phi}_E(t, t_1) e_{ij}(t_1) \tag{22}$$

on the consistent solution subspace, where $t_1 = \tau + T_s$.

b: Proof.

Subtract (20) for the two models. Every forcing term cancels by assumption, leaving the homogeneous descriptor variational equation. Applying its transition operator gives (22). □

When the retained PMU operator is locally affine, $y = H_y u + d$, the sampled response coefficients are

$$D_i = H_y s_i, \quad Q_i = \tfrac{1}{2} H_y s_{ii}, \quad Q_{ij} = H_y s_{ij}. \tag{23}$$

Thus all quadratic curvature comes from the physical solution and event map, not from PMU post-processing. This is why the dictionary must differentiate the same event semantics used to generate the trajectories.

C. EXACT FUNCTIONAL RECONSTRUCTION

Consider the finite-horizon linear experiment

$$Y = \mathcal{O}x + \varepsilon, \qquad \psi = Lx, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, R), \quad R \succ \mathbf{0}. \tag{24}$$

Theorem 3 (Functional reconstruction equivalence): The following statements are equivalent: (i) $\mathcal{O}x_1 = \mathcal{O}x_2$ implies $Lx_1 = Lx_2$; (ii) $\ker \mathcal{O} \subseteq \ker L$; and (iii) there is a linear map K such that $L = K\mathcal{O}$.

a: Proof.

If (i) holds and $v \in \ker \mathcal{O}$, the observations of v and $\mathbf{0}$ coincide, hence $Lv = \mathbf{0}$ and (ii) follows. If (ii) holds, define $K(\mathcal{O}x) = Lx$ on $\operatorname{range} \mathcal{O}$. The definition is independent of the representative because two representatives differ by a vector in $\ker \mathcal{O} \subseteq \ker L$. Extend K linearly to the observation space to obtain (iii). Finally, (iii) immediately implies (i). □

Let

$$G = \mathcal{O}^{\mathsf{T}} R^{-1} \mathcal{O}. \tag{25}$$

Theorem 4 (Best linear unbiased estimate of an estimable function): If the equivalent conditions of Theorem 3 hold, then

$$\widehat{\psi}_\star = LG^\dagger \mathcal{O}^{\mathsf{T}} R^{-1} Y, \tag{26}$$

$$\operatorname{Cov}(\widehat{\psi}_\star) = LG^\dagger L^{\mathsf{T}}. \tag{27}$$

It has minimum covariance, in the positive-semidefinite order, among all linear unbiased estimators of Lx [19].

b: Proof.

Because $v^{\mathsf{T}} G v = \|\mathcal{O}v\|_{R^{-1}}^2$, $\ker G = \ker \mathcal{O}$. Estimability gives $L = LG^\dagger G$, so taking expectation in (26) yields Lx. Direct covariance propagation gives (27). Let AY be any other linear unbiased estimator and set $T = A - LG^\dagger \mathcal{O}^{\mathsf{T}} R^{-1}$. Then $T\mathcal{O} = \mathbf{0}$ and the cross covariance vanishes because $LG^\dagger \mathcal{O}^{\mathsf{T}} T^{\mathsf{T}} = \mathbf{0}$. Consequently,

$$\operatorname{Cov}(AY) = \operatorname{Cov}(\widehat{\psi}_\star) + TRT^{\mathsf{T}} \succeq \operatorname{Cov}(\widehat{\psi}_\star). \tag{28}$$

If the kernel inclusion fails, two states induce the same data law but different targets, so no estimator can be unbiased for every state. □

The pseudoinverse has this interpretation only for estimable targets. In the nonlinear DAE, a pointwise kernel test is not sufficient: local factorization requires the condition to hold on a regular constant-rank neighborhood with connected local fibers [20].

D. FISHER INFORMATION WITH PHYSICAL NUISANCE

For a differentiable Gaussian model

$$r \mid \boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{\Sigma}(\boldsymbol{\theta})), \tag{29}$$

the Fisher information contains both mean and covariance contributions. For scalar a,

$$I(a) = \boldsymbol{\mu}_a^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_a + \tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}_a\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}_a). \tag{30}$$

The second term must be retained when event amplitude changes the covariance. The evaluated load bank freezes $\boldsymbol{\Sigma}$, so only the first term is used.

To expose nuisance geometry, consider

$$r = ad + N\boldsymbol{\nu} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{31}$$

where $\boldsymbol{\nu}$ may contain initial-state, operating-point, parameter, or integrity directions. Let $W^{\mathsf{T}}W = \boldsymbol{\Sigma}^{-1}$, $\overline{d} = Wd$, $\overline{N} = WN$, and

$$P_\perp = I - \overline{N}\overline{N}^{\dagger}. \tag{32}$$

Theorem 5 (Efficient Fisher information after nuisance profiling): For deterministic unknown $\boldsymbol{\nu}$, the efficient information for a is

$$I_{a|\nu}^{\mathrm{det}} = \overline{d}^{\mathsf{T}}P_\perp\overline{d} \tag{33}$$

$$= d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d - d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N(N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N)^{\dagger}N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d. \tag{34}$$

It is zero exactly when the event tangent lies in the observable nuisance span.

a: Proof.

After whitening, the score for a is $\overline{d}^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}}$. Nuisance scores span $\{v^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}} : v \in \operatorname{range}\overline{N}\}$. Their Fisher inner product is Euclidean because $\mathbb{E}[\overline{\boldsymbol{\varepsilon}}\overline{\boldsymbol{\varepsilon}}^{\mathsf{T}}] = I$. Removing the nuisance-score projection leaves $(P_\perp\overline{d})^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}}$, whose variance is (33). Expanding the projector by an SVD gives (34). □

If instead $\boldsymbol{\nu} \sim \mathcal{N}(m_\nu, \boldsymbol{\Gamma})$ with a proper prior, marginalization yields

$$r \mid a \sim \mathcal{N}(ad + Nm_\nu, \boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}}), \tag{35}$$

$$I_{a|\nu}^{\mathrm{Bayes}} = d^{\mathsf{T}}(\boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}})^{-1}d. \tag{36}$$

Woodbury's identity shows that (36) tends to the no-nuisance information as $\boldsymbol{\Gamma} \to \mathbf{0}$ and to (33) as a full-rank nuisance prior becomes diffuse. Thus zero deterministic efficient information is not a universal impossibility statement: a restrictive proper prior can still supply separation.

Corollary 1 (Pre-event information cannot reduce local event information): If pre-event data add nuisance precision $\boldsymbol{\Lambda}_{\mathrm{pre}} \succeq \mathbf{0}$, then

$$I_{a|\nu,\mathrm{pre}} = I_{aa} - I_{a\nu}(I_{\nu\nu} + \boldsymbol{\Lambda}_{\mathrm{pre}})^{-1}I_{\nu a} \tag{37}$$

is nondecreasing in $\boldsymbol{\Lambda}_{\mathrm{pre}}$.

b: Justification.

If $\boldsymbol{\Lambda}_2 \succeq \boldsymbol{\Lambda}_1 \succeq \mathbf{0}$, matrix inversion reverses the order. Premultiplication and postmultiplication by the cross-information blocks preserve the positive-semidefinite order, and subtraction reverses it once more. □

For a known source, known positive sign, and fixed covariance, the standardized matched statistic

$$T = \frac{d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}r}{\sqrt{d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d}} \tag{38}$$

is standard normal under H_0 and has mean $a\sqrt{I}$ under amplitude a. A level-α one-sided test has power $1-\Phi(z_{1-\alpha}-a\sqrt{I})$. Hence the amplitude required for power $1-\beta$ is

$$a_{\alpha,\beta} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{I}}. \tag{39}$$

This is a conditional known-source benchmark, not the threshold of the 137-support search. Its empirical use in the paper is limited to the exploratory source-wise detectability association.

E. FIRST-ORDER MULTI-EVENT SUPPORT GEOMETRY AND ONSET INFORMATION

Let two first-order source signatures be d_g and d_j under fixed precision $G \succeq \mathbf{0}$. Allow the competing source to refit its amplitude. The local profiled distance satisfies

$$b_\star(a) = a\frac{d_j^{\mathsf{T}}Gd_g}{d_j^{\mathsf{T}}Gd_j} + O(a^2), \tag{40}$$

$$\Delta^2_{g\to j}(a) = a^2 J_{g\to j} + O(a^3), \tag{41}$$

$$J_{g\to j} = d_g^{\mathsf{T}}Gd_g - \frac{(d_g^{\mathsf{T}}Gd_j)^2}{d_j^{\mathsf{T}}Gd_j} \geq 0. \tag{42}$$

Equation (42) is a pairwise local separation quantity, not the Fisher information for amplitudes conditional on the true support.

The pairwise formula extends to arbitrary supports. Let N_c be the unwhitened nuisance Jacobian, let $G_e = [g_1, \ldots, g_M]$ collect the event tangents, and define

$$A = WN_c, \quad F = WG_e, \quad R_N = I - AA^{\dagger}, \quad \overline{F} = R_N F. \tag{43}$$

Theorem 6 (Profile-likelihood diagnostic quotient): Consider the whitened local experiment

$$y = A\boldsymbol{\nu} + F\boldsymbol{\alpha} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I), \tag{44}$$

with deterministic unrestricted nuisance $\boldsymbol{\nu}$. Let $U_{\mathcal{D}}$ have orthonormal columns spanning $\mathcal{D} = \operatorname{range}(\overline{F})$, and define

$$z = U_{\mathcal{D}}^{\mathsf{T}}R_N y, \qquad d_{\mathcal{D}} = \dim\mathcal{D} = \operatorname{rank}(\overline{F}). \tag{45}$$

Then every profile-likelihood ratio over $\boldsymbol{\alpha}$ depends on y only through z:

$$\min_{\boldsymbol{\nu}} \|y - A\boldsymbol{\nu} - F\boldsymbol{\alpha}\|_2^2 = \|z - U_{\mathcal{D}}^{\mathsf{T}}\overline{F}\boldsymbol{\alpha}\|_2^2 + C(y), \tag{46}$$

where $C(y)$ is independent of $\boldsymbol{\alpha}$. The projector, quotient statistic up to an orthogonal change of coordinates, and $d_{\mathcal{D}}$ depend only on the nuisance span, not its parameterization.

a: Proof.

Profiling nuisance gives $\|R_N(y - F\boldsymbol{\alpha})\|_2^2$. Decompose this residual orthogonally into $\mathcal{D}$ and $\mathrm{range}(R_N) \cap \mathcal{D}^{\perp}$. The first component is the squared norm in (46); because $\overline{F}\boldsymbol{\alpha} \in \mathcal{D}$, the second is $C(y) = \|(I - U_{\mathcal{D}}U_{\mathcal{D}}^{\mathsf{T}})R_N y\|_2^2$ and contains no $\boldsymbol{\alpha}$. Reparameterizing nuisance without changing $\mathrm{range}(A)$ leaves $AA^{\dagger}$ and therefore every stated object unchanged. □

Thus $d_{\mathcal{D}}$ is the intrinsic dimension of the *linear profiled* event experiment. It is not a nonlinear global observability count and need not equal the number of PMUs. If candidates use different nuisance spans, or if nuisance has an informative proper prior, the corresponding candidate-specific profile or marginal likelihood must be retained; the common projector above is then insufficient.

Theorem 7 (Signed first-order support-to-support separation): For true support S with fixed severity direction $\boldsymbol{\alpha}_S$ and competing support C with freely fitted local amplitudes, define

$$\gamma_{S\to C}^2(\boldsymbol{\alpha}_S) = \min_{u,\boldsymbol{\beta}_C} \|F_S\boldsymbol{\alpha}_S - Au - F_C\boldsymbol{\beta}_C\|_2^2. \tag{47}$$

Then

$$\gamma_{S\to C}^2(\boldsymbol{\alpha}_S) = \|\left(I - P_{\overline{F}_C}\right)\overline{F}_S\boldsymbol{\alpha}_S\|_2^2, \tag{48}$$

$$= \boldsymbol{\alpha}_S^{\mathsf{T}} K_{S|C}\boldsymbol{\alpha}_S, \tag{49}$$

$$K_{S|C} = \overline{F}_S^{\mathsf{T}}\left(I - P_{\overline{F}_C}\right)\overline{F}_S \succeq \mathbf{0}. \tag{50}$$

Here $P_{\overline{F}_C} = \overline{F}_C\overline{F}_C^{\dagger}$, so rank deficiency in either nuisance or event coordinates does not change the projected trajectory.

b: Proof.

For fixed $\boldsymbol{\beta}_C$, minimizing (47) over u removes the component in $\mathrm{range}\,A$ and leaves $\|\overline{F}_S\boldsymbol{\alpha}_S - \overline{F}_C\boldsymbol{\beta}_C\|_2^2$. A second least-squares projection gives (48). Symmetry and idempotence of the orthogonal projector give (49) and its positive-semidefinite order. □

The theorem exposes two distinct weak-event mechanisms. For nonzero residualized single-source signatures $\overline{f}_i$ and $\overline{f}_j$,

$$\gamma_{i\to j}^2(a_i) = a_i^2\|\overline{f}_i\|_2^2(1 - \rho_{ij}^2), \qquad \rho_{ij} = \frac{\overline{f}_i^{\mathsf{T}}\overline{f}_j}{\|\overline{f}_i\|_2\|\overline{f}_j\|_2}. \tag{51}$$

Small $\|\overline{f}_i\|$ is event–nuisance invisibility; large $|\rho_{ij}|$ is event–event aliasing after nuisance removal.

For general supports, write $H = S \cap C$, $B = S \setminus C$, and $D = C \setminus S$. The shared amplitudes can be refitted, so

$$\gamma_{S\to C}^2 = \|\left(I - P_{[A,F_H,F_D]}\right)F_B\boldsymbol{\alpha}_B\|_2^2. \tag{52}$$

Thus the shared true component cancels exactly at first order for signed amplitudes or an interior parameter point. In particular, for $S = \{i,j\}$ and $C = \{i,k\}$ the relaxed margin is independent of the amplitude at i. Enlarging a competitor can only reduce the margin because its explanatory subspace grows.

The exclusive-support quotient removes the zero directions introduced by a literal support intersection. Define

$$\begin{aligned} Z_H &= [A, F_H], \quad R_H = I - Z_H Z_H^{\dagger}, \\ \widetilde{F}_B &= R_H F_B, \quad \widetilde{F}_D = R_H F_D. \end{aligned} \tag{53}$$

Then

$$\begin{aligned} \gamma_{S\to C}^2(\boldsymbol{\alpha}_B) &= \min_{\boldsymbol{\beta}_D} \|\widetilde{F}_B\boldsymbol{\alpha}_B - \widetilde{F}_D\boldsymbol{\beta}_D\|_2^2 \\ &= \boldsymbol{\alpha}_B^{\mathsf{T}} K_{B|D;H}\boldsymbol{\alpha}_B, \\ K_{B|D;H} &= \widetilde{F}_B^{\mathsf{T}}(I - P_{\widetilde{F}_D})\widetilde{F}_B \succeq \mathbf{0}. \end{aligned} \tag{54}$$

The first equality follows by minimizing first over the common span $\mathrm{range}\,Z_H$ and then over $\boldsymbol{\beta}_D$. Moreover, $K_{B|D;H} \succ \mathbf{0}$ if and only if

$$\ker \widetilde{F}_B = \{\mathbf{0}\}, \qquad \mathrm{range}\,\widetilde{F}_B \cap \mathrm{range}\,\widetilde{F}_D = \{\mathbf{0}\}. \tag{55}$$

Indeed, a zero quadratic form means that $\widetilde{F}_B\boldsymbol{\alpha}_B$ lies in the competing range; the two conditions force $\boldsymbol{\alpha}_B = \mathbf{0}$, and the converse is immediate. Under the coefficient normalization $\|\boldsymbol{\alpha}_B\|_2 = 1$, Rayleigh–Ritz gives the worst relaxed direction and margin as

$$\min_{\|\boldsymbol{\alpha}_B\|_2=1} \gamma_{S\to C}^2 = \lambda_{\min}(K_{B|D;H}). \tag{56}$$

This value depends on the physical scaling of the amplitude coordinates.

For a scale-free comparison, let Q_B, Q_D be orthonormal bases of the two conditional ranges. If $\theta_{\min}$ is their smallest principal angle,

$$\lambda_{\min}\left[Q_B^{\mathsf{T}}(I - Q_D Q_D^{\mathsf{T}})Q_B\right] = \sin^2\theta_{\min}. \tag{57}$$

The angle must be computed after removing nuisance and shared directions; otherwise every shared event creates a trivial zero angle. If the residual ambient dimension is $q_H = m - \mathrm{rank}\,Z_H$, then $r_B + r_D > q_H$ forces a nonzero intersection between the two conditional *spans*. This is a span-relaxed dimensional obstruction, not yet a one-sided physical ambiguity.

If $S \subset C$ and active amplitudes may equal zero, the competitor contains the true response and $\gamma_{S\to C} = 0$. If zero is excluded but belongs to the amplitude-domain closure, the infimum remains zero. Pure best-fit geometry therefore cannot select cardinality among nested support models. The marginal likelihood in Supplementary Appendix E supplies the required amplitude-volume and support-prior terms.

F. PHYSICAL TANGENT-CONE SEPARATION

Let $\mathcal{A}_C$ be the physical amplitude domain of a competing event model at the nominal point, and let $T_C = T_{\mathcal{A}_C}(\mathbf{0})$ denote its Bouligand tangent cone. Signed load changes give $T_C = \mathbb{R}^{|C|}$; one-sided trip, shedding, outage-continuation, and fault-severity coordinates give nonnegative rays. Missing PMU data are a discrete measurement-mode change and are not forced into this continuous construction.

Theorem 8 (Constrained first-order separation): Suppose a competing trajectory family has a local C^2 effective chart ϕ

with $\phi(\mathbf{0}) = y_0$, full-column-rank derivative J, and local parameter cone $K = \mathbb{R}^p \times T_C$, where T_C is closed and polyhedral. Assume no distant branch approaches y_0. For $y_\star(\epsilon) = y_0 + \epsilon b + o(\epsilon)$,

$$d_C^2(\epsilon) = \epsilon^2 \operatorname{dist}^2(b, JK) + o(\epsilon^2). \tag{58}$$

a: Proof.

For every $v \in K$, conicity gives $\phi(\epsilon v) = y_0 + \epsilon J v + o(\epsilon)$, which yields the upper bound after taking the infimum. Conversely, an $o(\epsilon)$-optimal local coordinate u_ϵ satisfies $\|\phi(u_\epsilon) - y_0\| = O(\epsilon)$. Full column rank and differentiability imply $\|u_\epsilon\| = O(\epsilon)$. Every convergent subsequence of u_ϵ/ϵ has a limit in the closed cone K, and division of the residual by ϵ gives the matching lower bound. The isolated-branch assumption prevents a nonlocal chart from attaining a smaller leading distance. □

With A the whitened unconstrained nuisance tangent and F_C the event tangent, nuisance profiling is still exact:

$$\gamma_{\text{phys}}^2 = \min_{\beta_C \in T_C} \|R_N(b - F_C\beta_C)\|_2^2 = \operatorname{dist}^2(R_N b, R_N F_C T_C). \tag{59}$$

The observable projection is unique even when the coefficient vector is not. Let $\mathcal{C} = R_N F_C T_C$ and $\mathcal{L} = \operatorname{span}\mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{L}$,

$$\gamma_{\text{phys}} \geq \gamma_{\text{span}}, \quad \gamma_{\text{phys}} = \gamma_{\text{span}} \iff \mathbf{\Pi}_{\mathcal{L}}(R_N b) \in \mathcal{C}. \tag{60}$$

Thus a positive span margin certifies physical separation, whereas a zero span margin does not prove one-sided ambiguity.

For $T_C = \mathbb{R}_+^{|C|}$, the optimum is a nonnegative least-squares projection. With residual $r = R_N(b - F_C\beta_C)$, the KKT conditions are

$$\beta_C \geq \mathbf{0}, \quad (R_N F_C)^\mathsf{T} r \leq \mathbf{0}, \quad \beta_j (R_N f_j)^\mathsf{T} r = 0. \tag{61}$$

For one true signature x and one admissible competing ray d,

$$\gamma_{\text{ray}}^2 = \|x\|_2^2 - \frac{[d^\mathsf{T} x]_+^2}{\|d\|_2^2}, \qquad [v]_+ = \max(0, v). \tag{62}$$

An antiparallel signature can therefore have zero span margin and a strictly positive physical margin.

For a shared one-sided support H, the exact difference coordinate satisfies $\delta_H \in T_H - \alpha_H$, which need not be a subspace. Hence the physical margin can depend on the shared true amplitude; the unconditional cancellation statement applies only when the relaxed solution is feasible. Competitor enrichment still cannot increase the margin whenever the old feasible set is contained in the new one. Likewise, a nested superset still has zero margin when it can match the shared amplitudes and set every added amplitude to zero.

For nontrivial closed cones $\mathcal{C}_S, \mathcal{C}_C$, the directed scale-free separation

$$\delta(\mathcal{C}_S \to \mathcal{C}_C) = \min_{\substack{x \in \mathcal{C}_S \\ \|x\|_2 = 1}} \operatorname{dist}(x, \mathcal{C}_C) \tag{63}$$

TABLE 3. Deterministic checks of conditional and one-sided geometry. These toy systems are not IEEE 39-bus evidence.

Check	Recomputed outcome
Conditional quotient	full–quotient error 0.0e+00 (pass).
Worst coefficient direction	eigenvalue–grid error 1.9e-08 (pass).
Conditional angle	naive 0.0e+00 deg; conditional 18.689 deg (pass).
Ray versus span	squared margins 0.0 versus 1.0 (pass).
One-sided shared source	squared-margin sequence 5.0, 2.0, 1.0, 1.0 (pass).
Cone enrichment	2000 trials; max violation 3.6e-15 (pass).
Nested cone	squared margin 2.4e-31 (pass).

vanishes exactly when the cones share a nonzero vector. It reduces to $\sin\theta_{\min}$ for subspaces, but need not be symmetric. Consequently, the classical signed-sparse condition $\operatorname{spark}(\overline{F}) > 2K$ remains sufficient under one-sided constraints, while its failure need not imply a physically feasible ambiguity.

For a multi-source support S, the Fisher information conditional on that known support is

$$I_S(a) = J_S(a)^\mathsf{T} G J_S(a), \quad J_S = \frac{\partial \boldsymbol{\mu}_S}{\partial a_S}. \tag{64}$$

Positive definiteness of I_S permits local amplitude estimation conditional on S; it does not exclude another support with the same mean.

Unknown onset has a separate degeneracy. For $\boldsymbol{\mu}(a, \tau) = a d(\tau)$ and fixed covariance,

$$I(a, \tau) = \begin{bmatrix} d^\mathsf{T}\mathbf{\Sigma}^{-1} d & a d^\mathsf{T}\mathbf{\Sigma}^{-1}\dot{d} \\ a\dot{d}^\mathsf{T}\mathbf{\Sigma}^{-1} d & a^2 \dot{d}^\mathsf{T}\mathbf{\Sigma}^{-1}\dot{d} \end{bmatrix}. \tag{65}$$

Profiling amplitude by a Schur complement gives

$$I_{\tau|a} = a^2 \left[\dot{d}^\mathsf{T}\mathbf{\Sigma}^{-1}\dot{d} - \frac{(d^\mathsf{T}\mathbf{\Sigma}^{-1}\dot{d})^2}{d^\mathsf{T}\mathbf{\Sigma}^{-1} d} \right]. \tag{66}$$

At $a = 0$, onset is unidentifiable. If $\dot{d}$ is proportional to d, delay and gain are locally confounded even for nonzero amplitude. The evaluated bank avoids this unresolved problem by fixing onset.

G. CANDIDATE INDISTINGUISHABILITY AND SPARSE INJECTIVITY

For candidate h and fixed integrity support S, define the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \operatorname{range}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \tag{67}$$

The profiled margin in the main paper is half the squared distance between two such sets under R_H^{-1}.

Theorem 9 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the candidate manifolds intersect. At any intersection, the two candidates induce the same Gaussian law; no decision rule can distinguish them uniformly over the declared nuisance set.

a: Proof.

Zero distance between closed affine sets means that there exist nuisance vectors for which their means coincide. The covariance is common by the stated contract, so the two conditional probability measures are identical. For any decision region, the probabilities of choosing i and j under this common law sum to one; at least one conditional error is therefore at least one half. □

For two fixed simple candidates with squared Mahalanobis distance $\Delta^2 = 2D_{ij,S}$ and equal priors, projection onto their mean difference reduces the optimal test to one dimension, giving

$$P_e^\star = \Phi(-\Delta/2) = \Phi(-\sqrt{D_{ij,S}/2}). \tag{68}$$

This formula does not apply directly to amplitude-marginalized support mixtures.

For the local persistent-event model $Y_T = \mathcal{H}_T a + \varepsilon_T$, define the physically admissible difference set

$$\mathcal{D}_K = \{a - b : a, b \text{ admissible and } K\text{-sparse}\}. \tag{69}$$

Theorem 10 (Finite-horizon sparse event uniqueness): Every admissible K-sparse event is uniquely determined in the noise-free local model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{D}_K = \{\mathbf{0}\}. \tag{70}$$

For a signed neighborhood in each support subspace, $\mathcal{D}_K = \mathcal{A}_{2K}$ locally, recovering the condition in the main paper.

b: Proof.

If two admissible events produce the same output, their nonzero difference belongs to both sets in (70). Conversely, any nonzero vector in the intersection is, by definition of $\mathcal{D}_K$, the difference of two admissible K-sparse events that produce the same output. □

Unknown initial-state nuisance can be removed locally by whitening and projection:

$$\widetilde{\mathcal{H}}_T = \left(I - P_{\mathrm{range}(W_T\mathcal{O}_T)}\right) W_T \mathcal{H}_T, \qquad W_T^\mathsf{T} W_T = R_T^{-1}. \tag{71}$$

Theorem 10 then applies with $\widetilde{\mathcal{H}}_T$. Define the restricted margin

$$\gamma_{2K}(T) = \min_{1 \le |U| \le 2K} \sigma_{\min}(\widetilde{\mathcal{H}}_{T,U}). \tag{72}$$

If two K-sparse explanations both fit data within whitened residual ϵ, then

$$\|a - b\|_2 \le \frac{2\epsilon}{\gamma_{2K}(T)}. \tag{73}$$

Indeed, the triangle inequality bounds $\|\widetilde{\mathcal{H}}_T(a-b)\|_2$ by 2ϵ, while the definition of γ_{2K} gives the reverse lower bound. For the evaluated $K \le 2$ bank, the relevant union size is four.

H. INFORMATION CANNOT DECREASE WHEN VALID MEASUREMENTS ARE ADDED

For a true trajectory at severity a and a competing physical family with parameter $\boldsymbol{\eta} \in \Theta_C$, define the exact profiled prefix distance

$$d_T^2(a) = \inf_{\boldsymbol{\eta} \in \Theta_C} r_T(a, \boldsymbol{\eta})^\mathsf{T} \boldsymbol{\Sigma}_T^{-1} r_T(a, \boldsymbol{\eta}), \tag{74}$$

where every $\boldsymbol{\Sigma}_T$ is a principal prefix of one positive-definite process covariance and the physical competitor domain does not change with T. Partition the difference between candidate means into old and added blocks, $\boldsymbol{\delta} = \mathrm{col}(\boldsymbol{\delta}_1, \boldsymbol{\delta}_2)$, with joint covariance $\boldsymbol{\Sigma} \succ \mathbf{0}$. Block Gaussian elimination gives

$$\boldsymbol{\delta}^\mathsf{T} \boldsymbol{\Sigma}^{-1} \boldsymbol{\delta} = \boldsymbol{\delta}_1^\mathsf{T} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\delta}_1 + \widetilde{\boldsymbol{\delta}}_2^\mathsf{T} \boldsymbol{\Sigma}_{2|1}^{-1} \widetilde{\boldsymbol{\delta}}_2, \tag{75}$$

$$\widetilde{\boldsymbol{\delta}}_2 = \boldsymbol{\delta}_2 - \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\delta}_1, \tag{76}$$

$$\boldsymbol{\Sigma}_{2|1} = \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12}. \tag{77}$$

Theorem 11 (Prefix monotonicity and temporal rescue): Under the prefix and fixed-domain contract above,

$$d_{T+1}^2(a) \ge d_T^2(a). \tag{78}$$

If both minima are attained, equality holds if and only if an element of $\operatorname{Argmin} d_T^2$ also has zero conditional innovation residual in the added block. Suppose further that, locally,

$$d_T^2(a) = c_T a^{2k_T} + o(a^{2k_T}), \qquad c_T > 0. \tag{79}$$

Then $k_{T+1} \le k_T$; if the order remains k, then $c_{T+1} \ge c_T$.

a: Proof.

The conditional covariance in (75) is positive definite, so its second quadratic form is nonnegative for every fixed $\boldsymbol{\eta}$. Taking the infimum over the same Θ_C proves (78). If the minima are attained, equality of the infima forces an extended-horizon minimizer to minimize the prefix term and to make the innovation term zero; the converse is immediate. If $k_{T+1} > k_T$, the ratio $d_{T+1}^2(a)/d_T^2(a)$ tends to zero as $a \to 0$, contradicting (78). When both orders equal k, division by a^{2k} and passage to the limit yield $c_{T+1} \ge c_T$. □

The theorem permits a longer prefix to reveal a first-order direction that was absorbed at an earlier horizon, or to reveal a higher-order obstruction that was previously invisible. It does not say that a concrete estimator improves: changing its normalization, nuisance set, covariance model, or optimization breaks the prefix contract. In particular, the archived RAWSIM39 calibration renormalized each event–signal–horizon block to unit weight and is not a prefix likelihood for this theorem.

If whitened model errors under two supports satisfy $\|We_S\| \le \epsilon_S$ and $\|We_R\| \le \epsilon_R$, the reverse triangle inequality gives the robust separation bound

$$d_{\mathrm{true}} \ge [d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+, \qquad \mathcal{J}_{\mathrm{true}} \ge \tfrac{1}{2}[d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+^2. \tag{80}$$

This is the required bridge for future simulator-shift claims; it is not evaluated by the current nominal load bank.

I. TEMPORAL INFORMATION UNDER THE FROZEN AR(1) CONTRACT

For true support S, competing support R, true amplitude a, and competitor amplitude b, let $\boldsymbol{\delta}_t(b) = \boldsymbol{\mu}_{S,t}(a) - \boldsymbol{\mu}_{R,t}(b)$. The frozen temporal covariance is

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}, \qquad [R_T]_{ij} = \rho^{|i-j|}, \quad |\rho| < 1. \tag{81}$$

For fixed b, its squared Mahalanobis separation has the innovation form

$$q_T(b) = \|\boldsymbol{\delta}_1(b)\|^2_{\boldsymbol{\Omega}^{-1}} + \frac{1}{1-\rho^2} \sum_{t=2}^{T} \|\boldsymbol{\delta}_t(b) - \rho\boldsymbol{\delta}_{t-1}(b)\|^2_{\boldsymbol{\Omega}^{-1}}. \tag{82}$$

Define the profiled simple-model separation

$$\mathcal{J}_{S\to R}(T) = \tfrac{1}{2} \inf_{b\in\mathcal{A}_R} q_T(b). \tag{83}$$

It equals the minimum KL divergence over fixed-amplitude simple Gaussian competitors. It is not the KL divergence or Bayes factor of the amplitude-mixture support model.

Theorem 12 (Persistent AR(1) information rate): If the amplitude domain is compact and fixed and $\boldsymbol{\delta}_t(b)$ converges uniformly to $\boldsymbol{\delta}_\infty(b)$, then

$$\lim_{T\to\infty} \frac{2\mathcal{J}_{S\to R}(T)}{T} = \frac{1-\rho}{1+\rho} \inf_{b\in\mathcal{A}_R} \|\boldsymbol{\delta}_\infty(b)\|^2_{\boldsymbol{\Omega}^{-1}}. \tag{84}$$

a: Proof.

Under uniform convergence, $\boldsymbol{\delta}_t - \rho\boldsymbol{\delta}_{t-1}$ converges uniformly to $(1-\rho)\boldsymbol{\delta}_\infty$. Divide (82) by T; the first-frame term vanishes and the Cesàro mean converges uniformly to $(1-\rho)^2(1-\rho^2)^{-1}\|\boldsymbol{\delta}_\infty\|^2_{\boldsymbol{\Omega}^{-1}}$. Compactness and uniform convergence permit interchange of infimum and limit, giving (84). □

Monotonicity of $\mathcal{J}_{S\to R}(T)$ follows independently from Theorem 11; the AR(1) calculation supplies the explicit limiting rate.

If the profiled equilibrium distance is positive, separation grows as $\Theta(T)$. If an admissible competitor matches equilibrium and its innovation differences are square summable, evaluating (83) at that competitor gives a finite upper bound; monotonicity then implies convergence to a finite ceiling. Exact trajectory overlap gives zero separation for every horizon.

For a true pair $\{i,j\}$, a competing pair $\{i,k\}$ can refit the shared source. With whitened equilibrium columns $\overline{h}_i, \overline{h}_j, \overline{h}_k$,

$$a_j^2\|(I - P_{i,k})\overline{h}_j\|_2^2 \le a_j^2\|(I - P_i)\overline{h}_j\|_2^2. \tag{85}$$

The inequality follows from $\operatorname{span}(\overline{h}_i) \subseteq \operatorname{span}(\overline{h}_i, \overline{h}_k)$. It formalizes why identifying which weak second source occurred cannot be easier than rejecting the shared single-source explanation.

J. HIGHER-ORDER CONTACT AND LOCAL DETECTION SCALE

First-order Fisher information can vanish while nonlinear separation remains. In whitened coordinates, let the no-event nuisance manifold and event curve be

$$F_0(b) = Nb + \tfrac{1}{2}H_0[b,b] + O(\|b\|^3), \tag{86}$$
$$c(a) = ad + a^2 q + O(a^3). \tag{87}$$

Theorem 13 (Relative first- and second-order separation): Assume the nuisance manifold has constant rank and use a regular chart after removing redundant coordinates. Define the profiled first-order coefficient

$$\gamma_1^2 = \|P_\perp d\|_2^2, \qquad P_\perp = I - (WJ_c)(WJ_c)^\dagger, \tag{88}$$

where J_c is the nuisance tangent before whitening and d is the already whitened event tangent in (87). If $\gamma_1 > 0$, the local squared distance to the nuisance manifold is

$$\Delta^2(a) = a^2\gamma_1^2 + O(a^3). \tag{89}$$

If $\gamma_1 = 0$, so the whitened event tangent lies in the nuisance tangent space, choose the regular-chart vector v satisfying $Nv = d$ and define

$$\boldsymbol{\kappa} = P_\perp\left(q - \tfrac{1}{2}H_0[v,v]\right). \tag{90}$$

Then

$$\Delta^2(a) = a^4\gamma_2^2 + O(a^5), \qquad \gamma_2^2 = \|\boldsymbol{\kappa}\|_2^2. \tag{91}$$

a: Proof.

The local minimizer satisfies $DF_0(b)^\mathsf{T}[F_0(b) - c(a)] = \mathbf{0}$. At the origin, its derivative with respect to the regular chart coordinate is $N^\mathsf{T}N \succ \mathbf{0}$, so the implicit-function theorem gives a smooth minimizing branch. Write $b_\star(a) = av + a^2 w + O(a^3)$. At first order, $N^\mathsf{T}(Nv - d) = \mathbf{0}$, yielding residual $aP_\perp d + O(a^2)$ and (89). If $d = Nv$, the first-order residual vanishes. At second order, minimization projects $q - \frac{1}{2}H_0[v,v]$ onto $\operatorname{range} N$; the remaining normal component is (90). Squaring the residual gives (91). The projector is unchanged by any full-rank reparameterization of the same nuisance tangent space, so γ_1 is a property of the local manifolds and covariance metric rather than of a particular nuisance coordinate basis. □

The same constant-rank qualification makes γ_2 intrinsic. Redundant original parameters may give several representatives of the same observable tangent, but their normal acceleration equals the second fundamental form of the same manifold branch. If $\gamma_2 = 0$, the conclusion is only $\Delta^2(a) = o(a^4)$ under the corresponding little-o expansion; an a^6 law or exact overlap requires a separate third-order analysis.

K. OPERATIONAL ERROR AND SUPPORT-RECOVERY LIMITS

For two simple hypotheses with a shared covariance, $P_i = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, define

$$d^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0). \tag{92}$$

The equal-prior log likelihood ratio is Gaussian with mean $\mp d^2/2$ and variance d^2 under P_0 and P_1, respectively. Hence

$$\begin{aligned} P_e^\star &= \Phi(-d/2), \quad D_{\mathrm{KL}}(P_1 \| P_0) = d^2/2, \\ C_{\mathrm{Ch}} &= d^2/8. \end{aligned} \tag{93}$$

The last identity gives $P_e^\star \le \frac{1}{2}\exp(-d^2/8)$. If a common affine nuisance is $A\boldsymbol{\eta}$, orthogonal projection after whitening removes it exactly and replaces d by the quotient margin. Thus $P_e^\star = \Phi(-\gamma/2)$ is exact for that quotient Gaussian experiment. It is not an exact formula for two nonlinear amplitude mixtures with different priors or dimensions.

Suppose the first nonzero contact order at horizon T is k and

$$d_T^2(a) = T\mathcal{I}_k |a|^{2k} [1 + o(1)], \qquad \mathcal{I}_k > 0. \tag{94}$$

For target simple-Gaussian error $\alpha < 1/2$, inversion of (93) yields

$$|a|_{\min}(\alpha, T) \sim \left[\frac{2\Phi^{-1}(1-\alpha)}{\sqrt{T\mathcal{I}_k}} \right]^{1/k}. \tag{95}$$

Equivalently, if $a_T = cT^{-\beta}$, the optimal binary error tends to one half for $\beta > 1/(2k)$, to a nontrivial constant at equality, and to zero for $\beta < 1/(2k)$. This gives the $T^{-1/2}$ and $T^{-1/4}$ boundaries for first- and second-order separation.

Now let $\Theta = \bigcup_{S \in \mathfrak{S}} \Theta_S$ and define the minimax exact-support risk

$$R^\star = \inf_{\widehat{S}} \sup_{\theta \in \Theta} \Pr_\theta\{\widehat{S} \neq S(\theta)\}. \tag{96}$$

Theorem 14 (Binary and minimum-severity limits): For every pair $\theta_i \in \Theta_{S_i}$ with $S_0 \neq S_1$,

$$\begin{aligned} R^\star &\ge \frac{1}{2}[1 - \mathrm{TV}(P_{\theta_0}, P_{\theta_1})] \\ &\ge \frac{1}{2}\left[1 - \sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(P_{\theta_0} \| P_{\theta_1})}\right]_+. \end{aligned} \tag{97}$$

If $S^+ = S \cup \{j\}$ and $P_{S^+, a_j} \to P_S$ in total variation as $a_j \downarrow 0$, then uniform exact-support recovery over all $a_j > 0$ has minimax risk at least one half.

a: Proof.

Restriction of any support rule to a two-point subset cannot reduce its worst-case risk; the optimal equal-prior binary risk is $[1 - \mathrm{TV}(P, Q)]/2$. Pinsker's inequality gives the second bound. For the nested pair, the binary risk tends to one half as the active amplitude tends to zero, so the supremum over positive amplitudes is at least that limit. □

The theorem requires either a declared $|a_j| \ge a_{\min} > 0$ contract or a decision that may abstain near absence. It also distinguishes a source-ranking error from an impossible exact-cardinality decision.

For M equiprobable supports, Fano's inequality gives

$$P_e \ge 1 - \frac{I(V;Y) + \log 2}{\log M}. \tag{98}$$

With $\overline{P} = M^{-1} \sum_j P_j$, convexity of KL in its second argument implies

$$\begin{aligned} I(V;Y) &= \frac{1}{M} \sum_i D_{\mathrm{KL}}(P_i \| \overline{P}) \\ &\le \frac{1}{M^2} \sum_{i,j} D_{\mathrm{KL}}(P_i \| P_j) = \overline{D}_{\mathrm{KL}}. \end{aligned} \tag{99}$$

Therefore a necessary condition for $P_e \le \alpha$ is

$$\overline{D}_{\mathrm{KL}} \ge (1 - \alpha) \log M - \log 2. \tag{100}$$

If $M = \binom{N}{K}$ and $\overline{D}_{\mathrm{KL}}(T) \le T\overline{\mathcal{I}}_k a^{2k}/2$, then necessarily

$$T \ge \frac{2[(1-\alpha)\log\binom{N}{K} - \log 2]}{\overline{\mathcal{I}}_k a^{2k}}. \tag{101}$$

This is an impossibility certificate, not a sufficient sample-size guarantee [21], [22]. For sparse $K \ll N$, it has the familiar scaling $T \gtrsim K \log(N/K)/(\overline{\mathcal{I}}_k a^{2k})$, now expressed through the physical Hybrid-DAE information rate.

L. DIAGNOSABILITY-PRESERVING DISCREPANCY COVARIANCE

A residual correction can improve fit while making physical events harder to distinguish. The following calculation makes that tradeoff explicit. Work after whitening and removal of common nuisance. Let $Q_U \in \mathbb{R}^{m \times r}$ be an orthonormal basis for the discriminant directions that must be protected, and let $Q_D \in \mathbb{R}^{m \times q}$ be an orthonormal discrepancy basis. The former should be frozen from critical candidate pairs; the latter must be estimated from mismatch-only records so that unlabeled events are not absorbed as model error.

Theorem 15 (Discrepancy retention and finite intensity): Let $P_D = Q_D Q_D^\top$ and model the whitened residual with covariance

$$C_s = I + sP_D, \qquad s = \tau^2 \ge 0. \tag{102}$$

Then $C_s^{-1} = I - \alpha P_D$, where $\alpha = s/(1+s)$, and the worst information fraction retained over the protected subspace is

$$\rho(s, Q_D) = 1 - \alpha\, \sigma_{\max}^2(Q_U^\top Q_D). \tag{103}$$

Set $c^2 = \sigma_{\max}^2(Q_U^\top Q_D)$ and $\delta = 1 - \rho_{\min}$. The constraint $\rho \ge \rho_{\min}$ imposes no finite upper bound on s when $c^2 \le \delta$; otherwise it requires

$$s \le s_{\max} = \frac{\delta}{c^2 - \delta}. \tag{104}$$

If the true residual covariance is $I + C_m$ and $E_D = \mathrm{tr}(Q_D^\top C_m Q_D)$, the expected Gaussian negative twice-log-likelihood relative to $s = 0$, up to constants, is

$$\Delta\mathcal{L}(s) = q \log(1+s) - \frac{s}{1+s}(q + E_D). \tag{105}$$

For fixed Q_D, its constrained minimizer is finite:

$$s^\star = \min\{E_D/q, s_{\max}\}, \tag{106}$$

with $s_{\max} = +\infty$ in the first geometric case.

TABLE 4. Deterministic checks for the discrepancy-retention identities. These are algebraic tests, not power-system experiments.

Check	Error	Tolerance	Status
Low-rank covariance inverse	4.53e-16	1.0e-12	PASS
Principal-angle retention	2.22e-16	1.0e-12	PASS
Retention variance cap	0.00e+00	1.0e-12	PASS
Finite full-likelihood intensity	5.50e-09	1.0e-07	PASS
Safe projected PCA	1.78e-15	1.0e-10	PASS
Protected-distance floor	1.11e-16	1.0e-12	PASS

a: Proof.

The inverse follows from $P_D^2 = P_D$. For $x = Q_U a$, the information ratio is the Rayleigh quotient of $I - \alpha Q_U^{\mathsf{T}} P_D Q_U$; its smallest eigenvalue is (103) because the singular values of $Q_U^{\mathsf{T}} Q_D$ are the principal-angle cosines. Solving $\alpha c^2 \le \delta$ gives (104). Finally, $\log|C_s| = q\log(1+s)$ and taking the expectation of $y^{\mathsf{T}} C_s^{-1} y$ under $\mathrm{Cov}(y) = I + C_m$ gives (105). Its derivative is $(qs - E_D)/(1+s)^2$, which gives (106). □

Under strict no-loss protection, $\rho_{\min} = 1$ and $s > 0$ require $\mathrm{range}(Q_D) \subseteq \mathrm{range}(Q_U)^{\perp}$. For fixed rank, the basis that captures the most mismatch energy is therefore the top-q eigenspace of $P_{U^{\perp}} C_m P_{U^{\perp}}$, provided that the orthogonal complement has dimension at least q. This is a safe projected-PCA baseline, not a learned-estimator result.

Maximizing only αE_D omits both the covariance determinant and the whitened measurement-noise contribution. It therefore drives s toward an infinite-variance limit whenever geometry permits. That utility is a valid description of reduced quadratic mismatch penalty, but it is not a normalized Bayesian tuning objective. The general joint orientation problem for $q > 1$ is an offline nonconvex design; spectral, semidefinite, or moment relaxations belong in a separate optimization study unless they are executed on the frozen event and mismatch dictionaries.

M. APPLIED GATE TAXONOMY FROM AN ARCHIVED ANDES CHECKPOINT

The frozen RAWSIM39 V1.2 calibration ledger provides an applied boundary check on the theory. It contains complete first-order screening and rank-selection decisions for 18 plant/event directions, but not the Jacobian and higher-order arrays needed to evaluate γ_2 or γ_3. Table 5 therefore classifies only what the recorded gates establish. In particular, failure of the curvature gate is not relabeled as second-order identifiability, and a small orthogonal fraction is not relabeled as exact redundancy.

N. THEORY-TO-EVIDENCE BOUNDARY

Equation (89) supplies a direct falsification test: for independently generated nonlinear trajectories with decreasing $|a|$, the profiled squared distance divided by a^2 should converge to γ_1^2. That convergence study has not been run for the present ANDES bank; the retrospective distance association and the gate-only RAWSIM39 taxonomy are not substitutes for it.

If n independent Gaussian replications have local distance $\Delta(a) = |a|^r \|k_r\| + O(|a|^{r+1})$, their KL divergence to the closest null is $n\Delta^2(a)/2$. Pinsker's inequality bounds the power of any level-α test by

$$\mathrm{Power}_a \le \alpha + \min\{1, \sqrt{n}\Delta(a)/2\}. \tag{107}$$

Thus $n|a_n|^{2r} \to 0$ prevents power from separating from size. The reference scales are $n^{-1/2}$ for first-order contact and $n^{-1/4}$ for second-order contact; singular-information asymptotics are established more broadly [23]. Correlated frames in one PMU window are not independent replications and must be handled through their joint covariance.

All audited load directions in the present bank have nonzero first-order separation. The second-order result is therefore a boundary theorem, not the empirical explanation for weak-pair failure.

TABLE 5. Conservative taxonomy of the frozen RAWSIM39 V1.2 calibration checkpoint. The gate sequence was 18 candidates, 10 individually eligible ($\kappa = 722.91$), eight rank-selected ($\kappa = 7.83$), and seven released after one 6.15σ local-domain rejection.

Category	Count	Directions	Evidence boundary
Regular local	7	plant_inertia_scale, plant_governor_droop_scale, generation_delta_p_pu, generation_delta_q_pu, generation_amplitude, generation_peak_s, load_delta_p_fraction	Released only under the local restricted posterior
Weak first-order	6	plant_damping, plant_governor_lag_scale, plant_turbine_time_scale, plant_avr_gain_scale, plant_avr_time_scale, plant_subtransient_x_scale	Low information; order and temporal rescue unresolved
Curvature unresolved	2	fault_duration_s, line_duration_s	Linear surrogate failed; not evidence of order 2
Near-redundant first-order	2	generation_tau_s, generation_residual	Small new orthogonal fraction; exact redundancy unproved
Local-domain failure	1	fault_reactance_pu	MAP left the declared local prior domain

TABLE 6. Recomputed checks of higher-order evidence and information limits. Controlled systems are not IEEE 39-bus performance evidence.

Check	Recomputed outcome
Evidence classes	slopes 0.9994, 0.4951, and 0.0003 for regular, order-2, and redundant additions (pass).
Prefix monotonicity	500 correlated-noise trials; max violation 0.0e+00 (pass).
Temporal rescue	first-order $\gamma_1^2(2) = 0.452724$; second-order relative error 1.0e-04 (pass).
Gaussian error map	quadrature error 5.6e-17 (pass).
Fano lower bound	$M = 8$ bounds 0.5993/0.3974; no Monte Carlo violation (pass).

TABLE 7. Deterministic checks of the local separation identities. These toy systems are not IEEE 39-bus evidence.

Check	Prediction and recomputed outcome
Regular first-order DAE	slope 2 and $\gamma_1^2 = 0.4527244641$; 2.000000; relative error 3.8e-12 (pass).
Shared-tangent second order	slope 4 and $\gamma_2^2 = 2.89$; 4.000000; coefficient 2.890000 (pass).
Rank-deficient nuisance	unique projected fit and $\gamma_1^2 = 0.5$; rank 2/3; max variation 1.2e-16 (pass).
Signed shared-source cancellation	margin independent of shared amplitude; squared-margin range 0.0e+00 (pass).
Competitor enrichment	$\gamma(S \to C_2) \leq \gamma(S \to C_1)$; 0.0196 versus 0.5096 (pass).
Nested support	geometric margin 0; 6.1e-35 (pass).
Principal angle	unit margin $\sin\theta$; absolute error 0.0e+00 (pass).
Nuisance enlargement	profiled margin cannot increase; sequence 2.0, 1.0, 0.0 (pass).
Diagnostic quotient	profiled costs factor through $d_{\mathcal{D}}$ coordinates; rank 4/6; max error 1.1e-15 (pass).
Nuisance reparameterization	unchanged nuisance span gives the same projector; spectral error 9.5e-16 (pass).

TABLE 8. Open checks for the simulator event semantics.

Check	Evidence required for closure
Right limit	Solver-scale $\|g_+(x(\tau+\epsilon), z(\tau+\epsilon), a)\|$ after re-equilibration; the callback residual may remain $O(a)$.
Algebraic derivatives	Agreement of analytic z_1, z_2 with central differences on a decreasing symmetric amplitude grid.
Taylor order	Constraint and state-approximation slopes $1, 2, 3$, or amplitude-halving ratios $2, 4, 8$.
Event families	One declared semantics and tolerance set for load, generation, and continuous branch-admittance perturbations.
Dynamic first order	Convergence of $\Delta y/a$ to $B_{r,a}$ over the declared early-time window.
Dynamic second order	Analytic second variation within a preregistered tolerance of nonlinear trajectories; the current discrepancy is unresolved.

TABLE 9. Status of the mathematical results relative to the frozen evidence.

Result	Present evidence boundary
Consistency jump	DAE contract only; callback/right-limit logging and derivative convergence remain open.
Functional equivalence and BLUE	Exact in the local linear model; nonlinear TEST accuracy remains empirical.
Fisher, support, and cone geometry	Signed and one-sided toy identities pass; IEEE 39-bus convergence and physical cone margins remain untested.
Sparse injectivity and γ_{2K}	T120 diagnostics concern the frozen load bank, not general descriptor left-invertibility.
Temporal order and rate	Prefix and controlled correlated-noise identities pass; the frozen long-horizon load study remains diagnostic and lacks independent operating-point validation.
Relative curvature	The controlled parabola passes; curvature-dominated weak-event behavior was not observed in the frozen atlas.
Error and minimax limits	Gaussian quotient, minimum-severity, and Fano formulas pass controlled checks; no IEEE 39 pairwise oracle-error atlas has been frozen.
RAWSIM39 gate taxonomy	The 18-direction ledger supports five conservative gate categories; derivative arrays needed to assign higher-order classes are unavailable.
Discrepancy retention	Exact for the whitened low-rank covariance model and checked algebraically; the protected dictionary, mismatch covariance, learned correction, and shifted IEEE 39-bus test remain unexecuted.

APPENDIX E ESTIMATOR AND LIKELIHOOD DETAILS

This appendix distinguishes a state-reconstruction component from two event estimator contracts. Sections E-A–E-C specify the state recursion and the implemented batch physical-manifold Bayes estimator $\mathcal{E}_{\mathrm{B}}$. Section E-F specifies the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. Filtering, smoothing, quadrature, and optimization are numerical operations within those contracts, not additional estimator proposals.

A. BAYESIAN RECURSION AND THE STATE HIERARCHY

For a Markov state model, Bayesian filtering alternates prediction and correction [24]. More broadly, this construction treats the PMU task as a Bayesian inverse problem in which physical candidate models and latent states are compared through predictive evidence [25]:

$$p(x_k \mid y_{0:k-1}) = \int p(x_k \mid x_{k-1}) p(x_{k-1} \mid y_{0:k-1})\, dx_{k-1}, \tag{108}$$

$$p(x_k \mid y_{0:k}) = \frac{p(y_k \mid x_k) p(x_k \mid y_{0:k-1})}{\int p(y_k \mid x_k) p(x_k \mid y_{0:k-1})\, dx_k}. \tag{109}$$

The denominator in (109) is the predictive evidence. Omitting it is harmless for a state mean under one fixed model, but not when physical candidates are compared.

B0 returns the equilibrium. B1 conditions the equilibrium prior on one PMU frame:

$$K_{\mathrm{LMMSE}} = P_0 C^{\mathsf{T}} (C P_0 C^{\mathsf{T}} + R)^{-1}, \tag{110}$$

$$\widehat{\delta x}_k^{\mathrm{B1}} = K_{\mathrm{LMMSE}} (y_k - y_0). \tag{111}$$

B2 retains state and covariance between frames. For the linear Gaussian model, substituting Gaussian densities into (108)–(109) yields

$$m_k^- = A_d m_{k-1}^+, \qquad P_k^- = A_d P_{k-1}^+ A_d^{\mathsf{T}} + Q, \tag{112}$$

$$S_k = C P_k^- C^{\mathsf{T}} + R, \qquad K_k = P_k^- C^{\mathsf{T}} S_k^{-1}, \tag{113}$$

$$m_k^+ = m_k^- + K_k (y_k - C m_k^-), \qquad P_k^+ = (I - K_k C) P_k^-. \tag{114}$$

Theorem 16 (Linear-Gaussian reduction): Equations (112)–(114) are the exact posterior mean and covariance generated by the Bayesian recursion for the stated linear Gaussian model.

a: Proof.

An affine transformation of the Gaussian posterior at $k-1$, followed by addition of independent Gaussian process noise, gives (112). The pair (x_k, y_k) is jointly Gaussian with cross covariance $P_k^- C^{\mathsf{T}}$ and measurement covariance S_k. The standard conditional-Gaussian formula gives (113)–(114). Equivalently, multiplying the predictive density by the likelihood and completing the square gives posterior precision $(P_k^-)^{-1} + C^{\mathsf{T}} R^{-1} C$; Woodbury's identity produces the same covariance and mean. □

The implementation may use the Joseph covariance form

$$P_k^+ = (I - K_k C) P_k^- (I - K_k C)^{\mathsf{T}} + K_k R K_k^{\mathsf{T}} \tag{115}$$

to preserve symmetry and positive semidefiniteness numerically.

B3 uses the Rauch–Tung–Striebel backward recursion [26]:

$$J_t = P_t^+ A_d^{\mathsf{T}} (P_{t+1}^-)^{-1}, \tag{116}$$

$$m_{t|k} = m_t^+ + J_t (m_{t+1|k} - m_{t+1}^-), \tag{117}$$

$$P_{t|k} = P_t^+ + J_t (P_{t+1|k} - P_{t+1}^-) J_t^{\mathsf{T}}. \tag{118}$$

It estimates a past state using later measurements. The negative B3 result therefore diagnoses model mismatch; it does not imply that correct future information generally lacks value.

The nominal uncertainty correction rescales the rectangular voltage covariance as

$$\boldsymbol{\Sigma}_{i,k}^{\mathrm{U2}} = T \boldsymbol{\Sigma}_{i,k} T^{\mathsf{T}}, \qquad T = \mathrm{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}), \tag{119}$$

with temperatures fitted on independent calibration trajectories. The observed-only adequacy statistic is a five-frame trailing mean of

$$a_k = (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu)^{\mathsf{T}} \boldsymbol{\Sigma}_\nu^{-1} (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu). \tag{120}$$

Neither operation changes the physical state estimate.

B. CONSTRAINED PHYSICAL RECENTERING

The static recentering estimator treats active/reactive load changes and active generator changes as nuisance coordinates d. It solves

$$(\widehat{V}, \widehat{d}) = \arg\min_{V,d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{\mathrm{AC}}(V,d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2. \tag{121}$$

Development data fix $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$. Because the AC constraint appears through a finite penalty rather than a normalized stochastic density, the implemented objective is a regularized physical estimator, not a fully normalized posterior.

For an exactly constrained local MAP, let $r(\boldsymbol{\chi})$ be the stacked residual, $c(\boldsymbol{\chi}) = \mathbf{0}$ the equality constraints, and $\boldsymbol{\Omega}$ the residual precision. A Gauss–Newton step solves

$$\begin{bmatrix} H & G_c^{\mathsf{T}} \\ G_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{\chi} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla J \\ -c \end{bmatrix}, \quad H = J_r^{\mathsf{T}} \boldsymbol{\Omega} J_r + H_{\mathrm{prior}}. \tag{122}$$

Theorem 17 (Feasible local covariance): If G_c has full row rank and Z spans $\ker G_c$, then $Z^\mathsf{T} H Z \succ \mathbf{0}$ makes the KKT matrix nonsingular. The local constrained Gaussian covariance is

$$P_\chi = Z(Z^\mathsf{T} H Z)^{-1} Z^\mathsf{T}. \tag{123}$$

a: Proof.

Any vector in the KKT nullspace satisfies $G_c v = \mathbf{0}$, hence $v = Zq$. Premultiplying the first block equation by v^T gives $q^\mathsf{T} Z^\mathsf{T} H Z q = 0$, so $q = \mathbf{0}$ and $v = \mathbf{0}$. Full row rank then forces the multiplier to zero. Parameterizing feasible perturbations by $\delta\boldsymbol{\chi} = Zq$ gives Gaussian precision $Z^\mathsf{T} H Z$ in q; its linear pushforward is (123). □

This covariance is local and unimodal. The reported joint NEES-like failure shows that it is not calibrated for the full shifted-plant experiment.

C. IMPLEMENTED BATCH PHYSICAL-MANIFOLD BAYES $\mathcal{E}_\mathsf{B}$

Let r be the centered T-frame PMU window. For support S with amplitudes a_S, the evaluated mean is

$$\boldsymbol{\mu}_S(a_S) = \sum_{i\in S}(a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j\in S\\ i<j}} a_i a_j Q_{ij}. \tag{124}$$

The W2 likelihood is

$$q_S(a_S) = \|r - \boldsymbol{\mu}_S(a_S)\|^2_{\boldsymbol{\Sigma}_T^{-1}}, \tag{125}$$

$$p(r \mid S, a_S) = \frac{\exp[-q_S(a_S)/2]}{(2\pi)^{n_y T/2}|\boldsymbol{\Sigma}_T|^{1/2}}, \tag{126}$$

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}. \tag{127}$$

The covariance is fitted only on normal calibration records and remains fixed across supports. The selected priors are

$$p(K = 0, 1, 2) = (0.20, 0.50, 0.30), \tag{128}$$

$$p(S \mid K) = \binom{16}{K}^{-1}, \tag{129}$$

$$a_i \mid i \in S \sim \mathcal{N}(0, \sigma_a^2), \qquad \sigma_a = 0.05. \tag{130}$$

For reference, the linear single-source model has a closed form.

Proposition 1 (Conjugate amplitude posterior and Bayes factor): Let $r = ad + \varepsilon$, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, and $a \sim \mathcal{N}(0, \sigma_a^2)$. Define $I = d^\mathsf{T}\boldsymbol{\Sigma}^{-1} d$ and $s = d^\mathsf{T}\boldsymbol{\Sigma}^{-1} r$. Then

$$v = (I + \sigma_a^{-2})^{-1}, \qquad m = vs, \tag{131}$$

$$a \mid r, S \sim \mathcal{N}(m, v), \tag{132}$$

$$\log \frac{p(r \mid S)}{p(r \mid H_0)} = \tfrac{1}{2}\frac{s^2}{I + \sigma_a^{-2}} - \tfrac{1}{2}\log(1 + \sigma_a^2 I). \tag{133}$$

a: Proof.

Terms in the joint log density that depend on a are $-\frac{1}{2}[(I + \sigma_a^{-2})a^2 - 2sa]$. Completing the square gives mean m and variance v. Integrating the resulting Gaussian contributes $\sqrt{v/\sigma_a^2}\exp(vs^2/2)$ relative to H_0; taking its logarithm gives (133). The logarithmic term is the amplitude-volume or Occam penalty. □

Corollary 2 (One-sided amplitude evidence): Under Proposition 1, replace the signed Gaussian slab by the proper half-normal prior $p(a) = 2\mathcal{N}(a; 0, \sigma_a^2)\mathbb{I}(a \geq 0)$. If $BF_\pm$ denotes the signed Bayes factor in (133), then

$$BF_+ = 2\Phi(z) BF_\pm, \qquad z = \frac{\sigma_a s}{\sqrt{1 + \sigma_a^2 I}}. \tag{134}$$

Conditional on the active model, the severity posterior is $\mathcal{N}(m, v)$ truncated to $[0, \infty)$.

b: Proof.

The fixed-amplitude likelihood ratio is $\exp(as - Ia^2/2)$. Multiplication by the half-normal density and completion of the square give twice the signed Gaussian integral restricted to $a \geq 0$. The unrestricted posterior has standardized probability $\Phi(m/\sqrt{v}) = \Phi(z)$ on that half-line, which yields (134). □

The factor $2\Phi(z)$ preserves direction: a large score with the prohibited sign is evidence against the one-sided event even though the signed factor depends on s^2. Proper slab scales are essential because Bayes factors across different support dimensions are undefined up to an arbitrary constant under an improper amplitude prior. Under the one-sided null $a = 0$, the likelihood ratio statistic satisfies

$$2\Delta\ell = [Z]_+^2, \qquad Z \sim \mathcal{N}(0, 1), \tag{135}$$

and therefore has the boundary mixture $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$, rather than the regular χ_1^2 law.

The quadratic multi-source mean is nonconjugate. Its evidence is

$$Z_S = p(r \mid S) = \int p(r \mid S, a_S) \prod_{i\in S} p(a_i)\, da_S. \tag{136}$$

With Q-node Gauss–Hermite abscissas x_q and weights w_q, the Gaussian-prior transformation $a_i = \sqrt{2}\sigma_a x_{q_i}$ gives

$$Z_S \approx \pi^{-|S|/2} \sum_{q_1=1}^{Q} \cdots \sum_{q_{|S|}=1}^{Q} \left(\prod_{j=1}^{|S|} w_{q_j}\right) p(r \mid S, \sqrt{2}\sigma_a x_q). \tag{137}$$

The implementation uses $Q = 31$ and evaluates (137) in the log domain using log-sum-exp. Independent adaptive Gauss–Kronrod integration checks representative two-dimensional cases. These checks support numerical integration under the frozen model; they do not validate the physical model itself.

D. POSTERIOR OUTPUTS AND THE CARDINALITY–LOCALIZATION DISTINCTION

The normalized support posterior is

$$p(S \mid r) = \frac{Z_S p(S)}{\sum_{R\in\mathcal{H}} Z_R p(R)}. \tag{138}$$

It induces cardinality, source-inclusion, and amplitude distributions:

$$p(K = k \mid r) = \sum_{|S|=k} p(S \mid r), \tag{139}$$

$$p(i \in S \mid r) = \sum_{S:\, i \in S} p(S \mid r), \tag{140}$$

$$p(a_i \mid r) = p(i \notin S \mid r)\delta_0 + \sum_{S:\, i \in S} p(S \mid r) p(a_i \mid S, r). \tag{141}$$

Equation (141) is a model-averaged spike-and-slab output: source absence contributes a point mass at zero.

The experimentally observed gap between cardinality and support has an exact algebraic decomposition. Let $N_k = \binom{16}{k}$, $\ell_S = \log Z_S$, and $\ell_k^\star = \max_{|S|=k} \ell_S$. Define the effective multiplicity

$$m_{\mathrm{eff},k} = \sum_{|S|=k} \exp(\ell_S - \ell_k^\star). \tag{142}$$

Proposition 2 (Exact effective-multiplicity identity): Under a uniform prior over supports of cardinality k,

$$\log p(r \mid K = k) = \ell_k^\star - \log N_k + \log m_{\mathrm{eff},k}, \tag{143}$$

$$\max_{|S|=k} p(S \mid K = k, r) = m_{\mathrm{eff},k}^{-1}. \tag{144}$$

a: Proof.

Factor $\exp(\ell_k^\star)$ from $N_k^{-1} \sum_{|S|=k} \exp(\ell_S)$ and take logarithms to obtain (143). Conditional normalization within cardinality k divides the largest evidence by the same sum, giving (144). □

Many similar pair supports can therefore increase evidence for $K = 2$ while preventing any one pair from dominating. Entropy separates the same two questions exactly because K is a deterministic function of S:

$$H(S \mid r) = H(K \mid r) + \sum_k p(K = k \mid r) H(S \mid K = k, r). \tag{145}$$

The first term measures cardinality uncertainty; the second measures localization ambiguity within each cardinality.

For set-valued output, sort supports by posterior mass and define the smallest credible set

$$\mathcal{C}_q(r) = \arg\min_{\mathcal{C}} |\mathcal{C}| \quad \text{subject to} \quad \sum_{S \in \mathcal{C}} p(S \mid r) \geq q. \tag{146}$$

Candidate-set size and coverage are more informative than an arbitrary Top-m when the posterior is multimodal. The present paper has not yet calibrated an abstention threshold or credible-set coverage prospectively.

E. STEP-BY-STEP EVALUATED BAYESIAN ALGORITHM

Table 10 records the exact order of the load-support computation. Steps marked "evaluated" produced the 137-hypothesis results; optional outputs are mathematically defined but are not promoted to validated claims.

The direct quadrature cost per record is

$$1 + GQ + \binom{G}{2} Q^2 \tag{147}$$

likelihood-node evaluations for $K \leq 2$. With $G = 16$ and $Q = 31$, this is 115,817 nodes before vectorization or reuse. The count explains why the support bank, integration order, and horizon must be frozen together when runtime is compared.

An incremental implementation of $\mathcal{E}_{\mathrm{B}}$ is computational, not a new estimator. If the exact innovations factorization includes the same initial AR(1) density, then at every frozen prefix it must reproduce the batch quadratic form, evidence, and posterior to numerical tolerance. This is a regression test, not evidence for the joint model below.

F. PROPOSED JOINT HYBRID-DAE BAYES $\mathcal{E}_{\mathrm{J}}$

The proposed extension would infer state trajectory X, algebraic variables Z, event family c, onset τ, support S, amplitudes a_S, operating nuisance $\boldsymbol{\eta}$, and integrity corruption b. Its posterior factorization is

$$\begin{aligned} p(X, Z, \boldsymbol{\eta}, b, c, \tau, S, a_S \mid Y) &\propto p(\boldsymbol{\eta}) p(c) p(\tau \mid c) p(S \mid c) \\ &\times p(a_S \mid S, c) p(\boldsymbol{\chi}_0 \mid \boldsymbol{\eta}) \\ &\times \prod_{t=0}^{T-1} p(\boldsymbol{\chi}_{t+1} \mid \boldsymbol{\chi}_t, \boldsymbol{\eta}, c, \tau, S, a_S) \\ &\times \prod_{t=0}^{T} p(y_t \mid \boldsymbol{\chi}_t, b_t, c, \tau, S, a_S) p(b_t), \end{aligned} \tag{148}$$

subject to the candidate DAE constraints. Here $\boldsymbol{\chi}_t$ denotes a consistent state/nuisance chart. A constrained negative twice-log density, excluding the discrete candidate priors, is

$$\begin{aligned} \mathcal{J}_S^\circ = \|\boldsymbol{\chi}_0 - \overline{\boldsymbol{\chi}}_0\|^2_{P_0^{-1}} + \sum_{t=0}^{T} \|r_{S,t}^{\mathrm{pmu}}\|^2_{R_t^{-1}} \\ + \sum_{t=0}^{T-1} \|r_{S,t}^{\mathrm{dyn}}\|^2_{Q_t^{-1}} + 2\lambda_b \sum_{t=0}^{T} \|b_t\|_1. \end{aligned} \tag{149}$$

Candidate evidence must be evaluated in a common feasible measure. The boundary correction is exposed by an explicit concentration parameter rather than by informally assuming a concentrated posterior.

Let $\mathcal{R}_0$ fix operating point, onset, and load family and replace the candidate prediction with the same quadratic manifold, W2 covariance, support prior, and amplitude prior used by $\mathcal{E}_{\mathrm{B}}$. Then the defining regression condition is

$$p_{\mathrm{J}}(S, a_S \mid Y, \mathcal{R}_0) = p_{\mathrm{B}}(S, a_S \mid r, \boldsymbol{\eta}_0, \tau_0, c_{\mathrm{load}}). \tag{150}$$

The equality is not claimed for an unrestricted nonlinear DAE likelihood; it holds only when the complete statistical contract is shared.

Theorem 18 (Local cone–Laplace evidence): Let $L_{S,\epsilon}$ be the log likelihood plus the log of a proper prior in identifiable

TABLE 10. Evaluated 137-hypothesis Bayesian load-support algorithm.

Step	Operation	Computation and frozen choice	Status
0	Declare contract	Fix 16 candidate load buses, $K \leq 2$, known onset, 30-frame window, eight PMUs, nominal topology, and no corrupted channels	Evaluated
1	Validate plant	Check equilibrium, terminal-current orientation, sampling, nonlinear solve acceptance, and first-order response before building the bank	Evaluated
2	Build physical dictionary	Generate centered D_i, self-curvature Q_i, and cross-interaction Q_{ij} from simulator-consistent interventions	Evaluated
3	Fit noise on CAL	Estimate channel covariance and AR(1) dependence from normal calibration records; freeze W2 before event TEST	Evaluated
4	Freeze priors and integrator	Set $p(K)$, uniform support priors, Gaussian amplitude slab, GH31 nodes/weights, and independent GK checks	Evaluated
5	Form the residual	Align the known onset, subtract the frozen normal mean, retain the fixed PMU/channel order, and apply the declared temporal covariance	Evaluated
6	Enumerate supports	Construct H_0, 16 single-source supports, and 120 two-source supports; no learned source label prunes the bank	Evaluated
7	Integrate each support	Evaluate (136) with (137); retain quadrature-weighted amplitude moments	Evaluated
8	Normalize once	Add the support prior in log space, subtract the global log-sum-exp, and obtain (138)	Evaluated
9	Read decisions in order	Report event presence, $p(K \mid r)$, conditional support ranking, source inclusion, and conditional amplitude intervals separately	Evaluated
10	Quantify ambiguity	Compute entropy, effective multiplicity, or $\mathcal{C}_q$ without relabeling them as validated abstention guarantees	Defined; partial evidence
11	Record provenance	Store the plant identifier, support, amplitude, noise seed, likelihood version, and frozen artifact hash for every independent record	Evaluated

local coordinates of dimension d_S. Suppose the rescaled feasible domains converge locally to a closed polyhedral cone K_S and

$$L_{S,\epsilon}(\boldsymbol{\xi}_0 + \epsilon u) - L_{S,\epsilon}(\boldsymbol{\xi}_0) = h_S^{\mathsf{T}} u - \tfrac{1}{2} u^{\mathsf{T}} H_S u + r_\epsilon(u), \quad (151)$$

where $H_S \succ \mathbf{0}$ and $r_\epsilon \to 0$ uniformly on compact sets. Assume an integrable dominating envelope and negligible evidence from distant modes. Then

$$Z_{S,\epsilon} = e^{L_{S,\epsilon}(\boldsymbol{\xi}_0)} \epsilon^{d_S} (2\pi)^{d_S/2} |H_S|^{-1/2} \times \exp\left(\tfrac{1}{2} h_S^{\mathsf{T}} H_S^{-1} h_S\right) \Pr(U_S \in K_S)[1 + o(1)], \quad (152)$$

$$U_S \sim \mathcal{N}(H_S^{-1} h_S, H_S^{-1}).$$

a: Proof.

Set $\boldsymbol{\xi} = \boldsymbol{\xi}_0 + \epsilon u$, which contributes the Jacobian ϵ^{d_S}. The compact-uniform expansion gives pointwise convergence of the rescaled integrand to $\exp(h_S^{\mathsf{T}} u - u^{\mathsf{T}} H_S u/2)$. The assumed envelope permits dominated convergence, while the distant-mode condition localizes the original integral. Completing the square in the limiting quadratic integral gives the Gaussian normalizer and the cone probability in (152). □

For an interior model $K_S = \mathbb{R}^{d_S}$ and the probability factor is one. For one or more boundary amplitudes it is a Gaussian orthant probability, not generally 2^{-q}. Half-normal prior normalizers supply their own factors in addition to this cone probability. Chart Jacobians, prior normalizers, and candidate-dependent covariance determinants belong in $L_{S,\epsilon}$ or the measure term. Comparing only optimized residuals drops both volume and boundary terms and does not produce a posterior.

For a small-noise Gaussian Hybrid-DAE trajectory map $\boldsymbol{\mu}_S(\boldsymbol{\xi}_0 + \epsilon u) = \boldsymbol{\mu}_0 + \epsilon J_S u + O(\epsilon^2)$,

$$h_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} z, \qquad H_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} J_S, \quad (153)$$

so positive curvature is equivalent to first-order identifiability in the reduced coordinates. This is a theoretical local specialization, not a validation of the unevaluated joint estimator. If H_S is singular, merely replacing its inverse by a pseudoinverse is invalid: redundant coordinates must first be quotiented out, while genuine higher-order identification changes the evidence scale. For example,

$$\int_0^\infty e^{-na^4/4}\, da = \frac{\Gamma(1/4)}{4^{3/4}} n^{-1/4}, \quad (154)$$

not the $n^{-1/2}$ rate of regular quadratic Laplace theory.

Proposition 3 (Exact reduction along redundant fibers): Suppose a parameter chart $\boldsymbol{\theta} \leftrightarrow (\boldsymbol{\xi}, \boldsymbol{\zeta})$ is one-to-one on the prior support and the observable map satisfies $\boldsymbol{\mu}(\boldsymbol{\theta}) = \overline{\boldsymbol{\mu}}(\boldsymbol{\xi})$ there. If $\widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})$ is the transformed proper prior, then the evidence equals the reduced integral with induced prior

$$\overline{\pi}(\boldsymbol{\xi}) = \int \widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})\, d\boldsymbol{\zeta}. \quad (155)$$

Thus an exactly redundant coordinate contributes no vanishing small-noise power; its effect is the pushforward prior measure, not a pseudodeterminant.

b: Proof.

The likelihood is independent of $\boldsymbol{\zeta}$. Fubini's theorem integrates the proper transformed prior along each fiber and gives $Z = \int p(y \mid \boldsymbol{\xi}) \overline{\pi}(\boldsymbol{\xi}) d\boldsymbol{\xi}$ exactly. □

For a merely local chart, the same reduction is asymptotic only after a localization condition excludes evidence from distant branches.

Theorem 19 (Mixed regular and higher-order evidence): In whitened coordinates, suppose the first observable expansion after redundancy removal is

$$\boldsymbol{\mu}(\boldsymbol{\xi}, a) = \boldsymbol{\mu}_0 + A\boldsymbol{\xi} + \frac{c_k}{k!} a^k + o(\|\boldsymbol{\xi}\| + |a|^k), \quad (156)$$

where A has rank r, the proper prior density is continuous and positive at the origin, and

$$c_{k,\perp} = (I - P_A) c_k \neq \mathbf{0}. \quad (157)$$

TABLE 11. Deterministic checks of boundary-aware evidence identities. These toy systems are not evaluated estimator results.

Check	Recomputed outcome
Signed/one-sided Bayes factors	max quadrature error 2.7e-12 (pass).
Broad nuisance limit	relative information error 2.1e-09 (pass).
Boundary likelihood ratio	zero mass 0.4972; positive mean 0.9987 (pass).
Nonlinear cone–Laplace	relative-error slope 0.9497 (pass).
Singular quartic evidence	scaling slope -0.2500 (pass).

Assume a dominating envelope and negligible evidence away from this branch. For the Gaussian-normalizer-free evidence kernel with noise scale ϵ,

$$M_\epsilon \sim C\epsilon^{r+1/k}, \tag{158}$$

$$C = \pi(\mathbf{0},0)(2\pi)^{r/2}|A^{\mathsf{T}}A|^{-1/2} \tag{159}$$

$$\times \frac{\Gamma(1/(2k))}{k}\left[\frac{2(k!)^2}{\|c_{k,\perp}\|_2^2}\right]^{1/(2k)}. \tag{160}$$

c: Proof.

Use $\boldsymbol{\xi} = \epsilon u$ and $a = \epsilon^{1/k}t$. The Jacobian is $\epsilon^{r+1/k}$ and the limiting kernel is $\exp[-\|Au + c_k t^k/k!\|^2/2]$. The component of c_k in $\operatorname{range} A$ disappears after a translation of u. Gaussian integration over the remaining regular coordinates yields $(2\pi)^{r/2}|A^{\mathsf{T}}A|^{-1/2}$; integration of the normal component over $t \in \mathbb{R}$ gives the gamma-function factor. Dominated convergence supplies the prior density at the origin and the stated limit. □

For one additional amplitude, the theorem separates three cardinality cases. A regular direction supplies an $O(\epsilon)$ Occam factor, a direction first visible at order k supplies $O(\epsilon^{1/k})$, and an exactly redundant extension supplies an $O(1)$ factor determined by its induced prior. For $\epsilon \asymp n^{-1/2}$ these become $n^{-1/2}$, $n^{-1/(2k)}$, and n^0. These are local evidence rates; the RAWSIM39 gate ledger does not contain the higher-order derivatives required to assign them to its rejected parameters.

For a sliding window, eliminating old Gaussian variables creates an arrival prior. Write its canonical log kernel as $-\frac{1}{2}z^{\mathsf{T}}\Lambda z + h^{\mathsf{T}}z + c$. If the information matrix and vector are partitioned into variables a to eliminate and retained variables b, exact integration yields

$$\Lambda_{\text{new}} = \Lambda_{bb} - \Lambda_{ba}\Lambda_{aa}^{-1}\Lambda_{ab}, \tag{161}$$

$$h_{\text{new}} = h_b - \Lambda_{ba}\Lambda_{aa}^{-1}h_a, \tag{162}$$

$$c_{\text{new}} = c + \tfrac{1}{2}h_a^{\mathsf{T}}\Lambda_{aa}^{-1}h_a + \tfrac{n_a}{2}\log(2\pi) - \tfrac{1}{2}\log\det\Lambda_{aa}. \tag{163}$$

The first two lines preserve the next MAP problem; the last line is also required to preserve evidence between candidates.

Table 12 gives the intended sequence. It is a specification for the next experiment, not a claim about the results reported in the main paper.

For a posterior state $\boldsymbol{\pi}_t(S) = p(S \mid Y_{1:t})$, an idealized unit-cost support decision with per-frame delay cost c satisfies the Bellman equation

$$V(\boldsymbol{\pi}) = \min\left\{1 - \max_S \pi(S),\ c + \mathbb{E}[V(\boldsymbol{\pi}_{t+1}) \mid \boldsymbol{\pi}_t = \boldsymbol{\pi}]\right\}. \tag{164}$$

The one-step expected information gain is

$$I(S; y_{t+1} \mid Y_{1:t}) = H(S \mid Y_{1:t}) - \mathbb{E}[H(S \mid Y_{1:t+1})]. \tag{165}$$

Equations (164)–(165) motivate adaptive horizon selection. They are not an executed stopping rule for the current fixed-onset, fixed-window bank.

Finally, a hidden-voltage or derived-output posterior must average across support uncertainty. With $w_S = p(S \mid Y)$, conditional output mean m_S, and covariance P_S,

$$\overline{m} = \sum_S w_S m_S, \tag{166}$$

$$\operatorname{Cov}(\psi \mid Y) = \sum_S w_S P_S + \sum_S w_S (m_S - \overline{m})(m_S - \overline{m})^{\mathsf{T}}. \tag{167}$$

The second term is between-support ambiguity. Nonlinear output moments may be computed from posterior samples or a local delta method; phasors are averaged in rectangular coordinates before magnitude and angle are formed.

G. ESTIMATOR CLAIM BOUNDARY

The state recursion is an independently evaluated component. $\mathcal{E}_{\mathrm{B}}$ computes normalized Bayesian evidence under the frozen quadratic W2/L2 model and is the implemented reference event estimator. $\mathcal{E}_{\mathrm{J}}$, its reduction test, learned discrepancy, unknown-onset/family inference, changing-PMU operation, virtual outputs, and sequential stopping remain specified but unexecuted. The physical recentering is a regularized constrained estimator with separately assessed uncertainty; it is not relabeled as either event estimator.

APPENDIX F COMPLETE FROZEN EXPERIMENTAL CONTRACTS

The PowerDynamics validation ladder checks static translation, PMU terminal-current orientation, equilibrium residual, local modes, nonlinear acceptance, and first-order response before estimator scoring. All 980 main mismatch cases and 175 refinement cases pass the declared acceptance gates. The physical-recentering TEST includes 20 equilibria at each of three shift scales; a solve is accepted only when the optimizer succeeds, outputs are finite, and the maximum AC residual is at most 5×10^{-5}. Rejections remain in the denominator.

For GLOBAL-137, the physical dictionary, covariance, priors, onset, and quadrature are frozen before new trajectories are generated. Three disjoint noise realizations accompany each event path, and 100 additional H_0 records represent no event. The identifiers have zero overlap with 1,816 historical records. The T120 bank preserves exact prefixes at 30, 60, 90, and 120 frames; independent operating-point validation remains pending.

TABLE 12. Candidate-conditioned joint estimator design and current implementation boundary.

Step	Operation	Required calculation	Present status
1	Pre-event update	Use only pre-event measurements to update state and operating nuisance; retain cross covariance and normalized arrival evidence	State filter evaluated; joint nuisance pending
2	Integrity mask	Mark missing channels exactly and introduce sparse corruption variables for available but suspect channels	Ontology defined; common inference pending
3	Candidate generation	Enumerate physical family, support, onset, and severity domain without using source identity as a learned label	Load family evaluated; other families pending
4	Candidate prediction	Apply the candidate DAE/event map and the current PMU membership to generate a physical trajectory	Load bank evaluated
5	Candidate-conditioned solve	Fit state and nuisance on the feasible DAE manifold, conditional on the physical candidate and amplitude quadrature node	Not executed jointly
6	Learned discrepancy	If used, predict only candidate-residual discrepancy and covariance; do not replace the physical hypothesis or consume target source labels	Not implemented
7	Evidence normalization	Retain GH31 amplitude marginalization when $\lvert S\rvert \leq 2$; integrate or approximate the remaining coordinates with all determinants, priors, and Schur normalizers	GH31 load evidence evaluated; joint evidence pending
8	Posterior outputs	Return state-functional mixture, cardinality, support, integrity state, severity, and candidate-set uncertainty	Separate state/load outputs only
9	Sequential decision	Continue observing when expected information gain justifies delay; use distinct thresholds for detection, cardinality, and exact source	Not evaluated
10	Reconfigure and verify	After accepting a physical change, rebuild the operating center and check subsequent innovations before declaring a persistent model update	Static recentering only

TABLE 13. Complete frozen-contract inventory. The main paper groups these contracts by research question.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families and seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source-amplitude-noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; weak and finite TEST blocks	Source-amplitude-noise record
Prospective 137-hypothesis inference	Fresh H_0 + 16 singles +120 pairs; 3,488 paths; 10,564 decisions	Physical trajectory or complete record, as declared
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 paths	Retrospective source-pair trajectory
Long-horizon diagnostic	Exact prefixes through 120 frames; 32 nominal pair paths; 137 hypotheses	Diagnostic trajectory, not population recovery
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied	101 event parents, each evaluated across three seeds

The ANDES ordinary split uses three replicas for training, one for development, and one for testing at each source configuration. The source-holdout audit removes the physical asset from every source-labeled operation and supplies the true event family at evaluation. These contracts answer interpolation and source generalization separately.

APPENDIX G SECONDARY RESULTS

The main paper retains the figures needed for the scientific argument. This appendix records secondary diagnostics and exact values that would otherwise interrupt that argument. The two compact tables first preserve the exact mismatch and weak-signal values; the full-width panels then show the corresponding temporal, likelihood, and supervised-baseline diagnostics.

TABLE 14. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

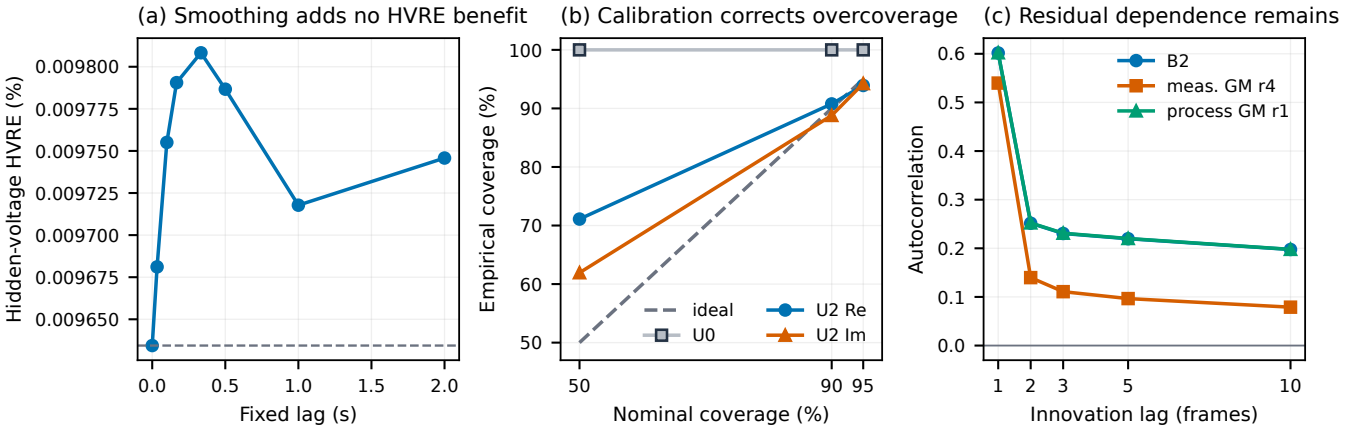


FIGURE 1. **Secondary temporal and uncertainty diagnostics for B2/B3 and calibration variants. These panels diagnose model adequacy; they do not upgrade nominal marginal coverage to a complete uncertainty claim.**

TABLE 15. **Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.**

$\lvert a \rvert$ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

TABLE 16. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

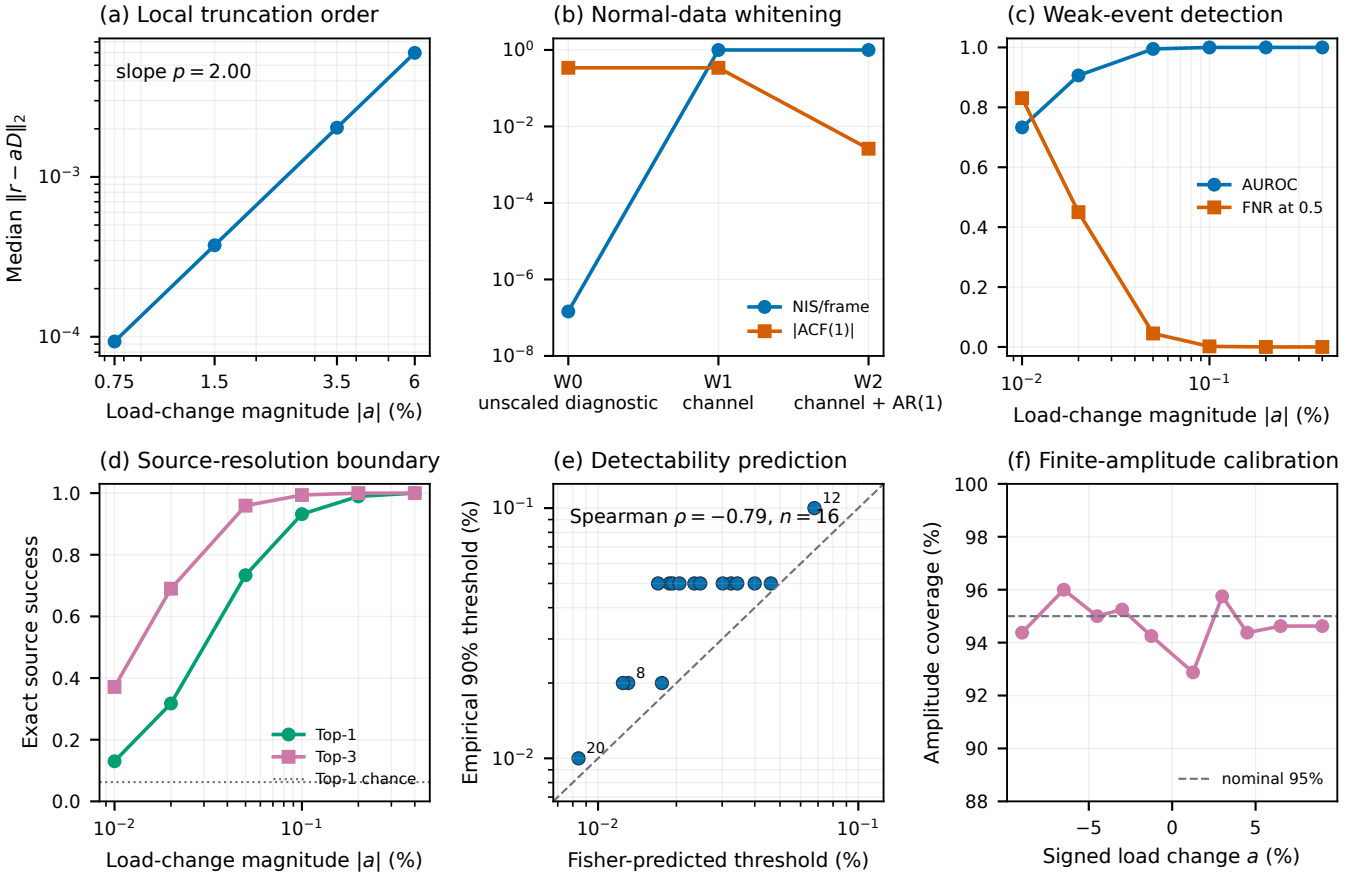


FIGURE 2. Single-source development evidence for the frozen load likelihood. The figure is retained as component validation; GLOBAL-137 in the main paper is the prospective support-recovery result.

TABLE 17. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

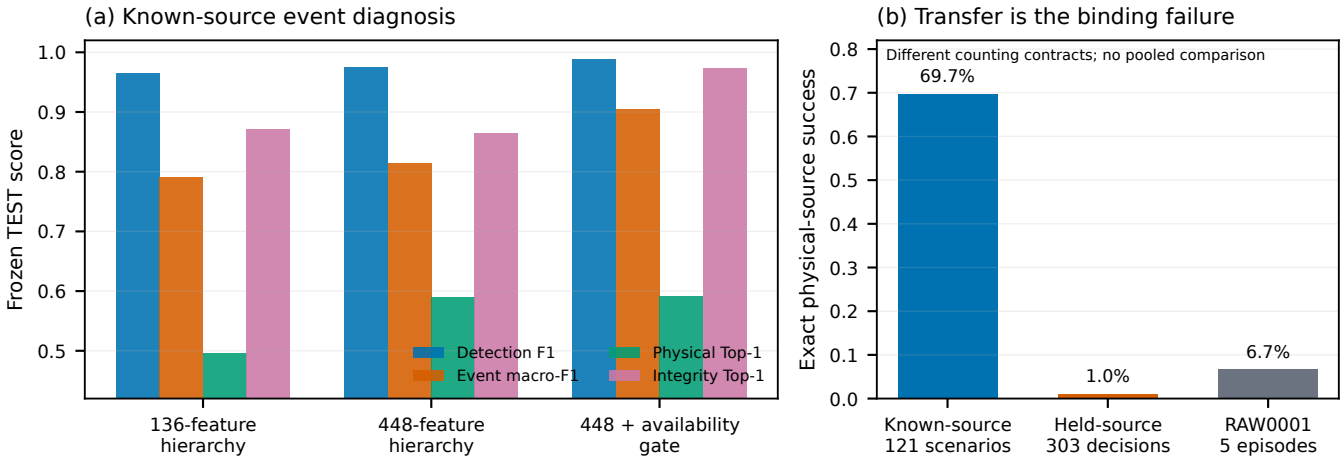


FIGURE 3. **Secondary evidence for the supervised ANDES baseline, including known-source performance and the source-holdout failure.**

APPENDIX H EXPLORATORY HIERARCHICAL SINGLE-EVENT PILOT

This appendix records the September 24, 2026 estimator handoff as a *provisional* numerical branch of the joint design $\mathcal{E}_{\mathrm{J}}$. Its source documents are the 30-page mathematical derivation book and 24-page technical integration dossier identified in the repository integration ledger. Their statements are technical source material, not independent instructions or substitutes for campaign outputs. The dossiers label many checks "artifact-verified" in packages supplied elsewhere; those packages were not supplied with this manuscript checkout. The latest strict-final comparison is explicitly tagged terminal-reported in both documents. All values below are therefore transcribed as *dossier-reported* diagnostics and excluded from the article's frozen primary tables.

A. STATE, MEASUREMENTS, AND EVENT OPERATORS

The pilot's declared model has 114 differential states, 78 algebraic rectangular-voltage coordinates, and 32 measured rectangular channels: voltage at eight PMUs and terminal current at eight monitored branches. Sampling is at $1/30$ s. Its default matrices are $P_0 = 10^{-2} I_{114}$, $Q = 10^{-6} I_{114}$, and $R = 10^{-6} I_{32}$. These numbers identify the pilot likelihood, not the separately frozen covariance W2 of $\mathcal{E}_{\mathrm{B}}$. Three-phase phasors, frequency, ROCOF, and data-present fields in the historical competition files do not silently become inputs to this 32-channel physical likelihood.

For a fixed candidate H and severity ϑ_H, the semi-explicit system is $\dot{x} = f_H(x, z; \vartheta_H)$, $\mathbf{0} = g_H(x, z; \vartheta_H)$, and $y = h_H(x, z) + v$. At a regular point with invertible $G_z = \partial g_H / \partial z$, the implicit-function theorem gives

$$Z_x = -G_z^{-1} G_x, \quad Z_\vartheta = -G_z^{-1} G_\vartheta, \quad A_c = F_x - F_z G_z^{-1} G_x. \tag{168}$$

The corresponding reduced event pathway is $B_\vartheta = F_\vartheta + F_z Z_\vartheta$ and the direct measured pathway is $D_\vartheta = H_\vartheta + H_z Z_\vartheta$. A generator-reference change can satisfy $D_\vartheta = \mathbf{0}$ while $B_\vartheta \neq \mathbf{0}$: it is invisible to a raw algebraic screen but visible after dynamic propagation. The event operator and its duration must be replayed consistently in truth generation and candidate fitting. The pilot dossier reports a five-frame fault pulse, persistent load steps unless recovery is declared, and finite generator and line events with recovery. Some earlier inference runs used persistent approximations for the latter two; those rankings have a known model-contract discrepancy.

The current single-event catalog has 39 fault buses, 19 load buses, nine controlled generators, 46 in-service branches, and a separate H_0 comparison. Thus it has 113 physical candidates and one null candidate; the screening rank denominators may be smaller when a candidate is numerically unscorable. Fault severity uses a log-resistance coordinate, load severity two active/reactive changes, generator severity a reference-power change, and an outage has a discrete branch and switching profile. Missing or bad PMU channels belong to the integrity variable of $\mathcal{E}_{\mathrm{J}}$ and were not jointly inferred in this pilot.

B. JOINT GAUSS–NEWTON STEP AND ITS INTERPRETATION

For one candidate, write the conditional MAP residual as $r_H(X, \vartheta)$ after whitening and appending proper prior residuals. Let $\Delta u = \mathrm{col}(\Delta X, \Delta\vartheta)$ and $g = \mathrm{col}(g_X, g_\vartheta)$. At an iterate, its local quadratic is

$$q(\Delta u) = g^{\mathsf{T}} \Delta u + \tfrac{1}{2} \Delta u^{\mathsf{T}} H \Delta u, \qquad H = \begin{bmatrix} H_{XX} & H_{X\vartheta} \\ H_{\vartheta X} & H_{\vartheta\vartheta} \end{bmatrix}. \tag{169}$$

Eliminating ΔX gives $q_{\mathrm{red}}(\Delta\vartheta) = \mathrm{const} + b_H^{\mathsf{T}} \Delta\vartheta + \tfrac{1}{2} \Delta\vartheta^{\mathsf{T}} S_H \Delta\vartheta$, where $S_H = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ and $b_H = g_\vartheta - H_{\vartheta X} H_{XX}^{-1} g_X$. Hence its unconstrained local decrease is $-\tfrac{1}{2} b_H^{\mathsf{T}} S_H^{-1} b_H$ when $S_H \succ 0$. This algebra needs no conditional convergence of X: it is the full-block Gauss–Newton step expressed with a smaller parameter solve. By contrast, if $\psi(\vartheta) = \min_X J_H(X, \vartheta)$, the identity $\nabla^2 \psi = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ is a local *profile-Hessian* statement only at a regular conditional optimum, and for the true Hessian rather than an arbitrary Gauss–Newton approximation. An unsettled state trajectory can change the sign of the reduced gradient without invalidating Schur elimination.

The numerical realization factors the temporally sparse state block and obtains its action on g_X and the few columns of $H_{X\vartheta}$ by solves; it does not form H_{XX}^{-1}. One Armijo step length updates both state and severity and evaluates the actual nonlinear DAE objective. The pilot selected Rodas4P for transition evaluation after the reported cross-integrator audit. That choice is a numerical guard, not a new inference theorem. Levenberg–Marquardt remains an independent finalizer candidate for the unresolved fault/line fits [27], [28].

C. DETECTION, PHYSICAL SCREENING, AND RACING

Let $\boldsymbol{\nu}_k = y_k - \widehat{y}_{k|k-1}^{H_0}$ be the observed innovation. An alarm tests the adequacy of H_0 on measured channels; it is not itself the physical onset. The pilot refines onset over a bounded interval and fits the common initial state $\widehat{X}_{\mathrm{pre}}$ on data before that onset. Every candidate starts from this same reference. This prevents a strong event from contaminating its own null-model baseline.

For a smooth candidate nested at H_0, the efficient local score is $T_H = b_H^{\mathsf{T}} S_H^{-1} b_H$ under positive definite S_H. If a severity is physically one-sided, maximizing the local quadratic over its tangent cone replaces the unconstrained inverse step and changes the null law; the ordinary signed score is then only a proposal proxy. For a fault or topology switch, let s_H be its short-horizon physical response from $\widehat{X}_{\mathrm{pre}}$ and r the corresponding null residual. With $W^{\mathsf{T}} W = R_H^{-1}$ and nuisance projector P_N, define $\widetilde{s}_H = (I - P_N) W s_H$ and $\widetilde{r} = (I - P_N) W r$. A signed matched score is $(\widetilde{r}^{\mathsf{T}} \widetilde{s}_H)^2 / \|\widetilde{s}_H\|^2$, with the allowed severity domain imposed before interpreting it as a physical fit. The two screening scores need not share a calibrated scale across families.

The frozen pilot consequently ranks within each family, retains a family quota, and allocates increasing nonlinear

TABLE 18. Single-event pilot ledger transcribed from the two author-supplied September 24 dossiers. All rows require the corresponding campaign archives for an independent paper audit.

Check or campaign	Dossier-reported result and precise interpretation	Source tier
Linear Gaussian oracle E0	RTS and batch MAP means agree near 10^{-16}; covariance/log-likelihood checks near 10^{-14} after an omitted evidence constant was repaired. This guards the numerical algebra only.	Dossier AV
Three-node and five-bus toys	E1: five development and 30 evaluation seeds, hidden algebraic two-sigma coverage 0.927 versus nominal 0.954. E2: first-order classification 0.2875 over 10 hypotheses × eight repetitions; not a nonlinear full-catalog result.	Dossier AV
Transition/derivative audit	Rodas4P/Rodas5P: 23 shared successful candidates with maximum endpoint difference 4.9×10^{-7}; Rodas4P evaluated three of three selected Rodas5P failures. Finite-difference coverage was 25/25; one random direction retained a moderate derivative discrepancy.	Dossier AV
Joint update and severity	Full-block and Schur steps agree in 30/30 synthetic block systems to numerical precision. Four fault-resistance initializations of 0.01, 0.03, 0.05, and 0.08 ended in 0.02006–0.02133 around truth 0.021; only the first two passed the stated strict stationarity gate.	Dossier AV
Small hidden-event bank	A hidden LOAD@16 and LINE(23,24) ranked first against selected decoys after joint severity and line-index corrections. These were not full-catalog tests.	Dossier AV
H_0 and onset pilots	Fifty-one disjoint normal calibration windows yielded no alarms at two reported thresholds on that same set; this is neither independent alarm testing nor false alarms per hour. Blind onset alternatives differed by 26–27 frames in two cases.	Dossier AV
Stage-A development	Four cases; 451/456 attempted physical scores were valid (98.9%); family-stratified shortlist retained 4/4 truths.	Dossier AV
Blind mini-holdout	FAULT@20, LOAD@28, GEN@38, and LINE(4,14): within-family truth ranks 3/38, 1/19, 1/9, and 1/46. The shortlist and subsequent racing retained 4/4 truths; GEN survived at the six-candidate cutoff. Four quiet windows were a proposal-level control only.	Dossier AV
Strict final comparison	The later terminal summary reports 0/4 exact winners after 24 final fits. Fault and line include unconverged truth/competitors; load and generator have better-scoring converged neighbors. This is a failure audit, not an identifiable physical error rate.	TR; final archive absent

budgets $B1, B2, B3, B5, B10$ before strict finalization. A candidate with a failed score is recorded as unscorable rather than inserted into a tie by catalog order. The dossier reports that ranking was nonmonotone with iteration budget and that the generator truth survived the blind racing cutoff at rank six of six. Thus survival is a search diagnostic; it says nothing by itself about a final converged winner or computational savings. In the reported racing run all 21 candidates were still evaluated to ten iterations, so a wall-clock advantage has not been demonstrated.

The operational order of this *pilot* is fixed: (1) process only the declared 32-channel PMU sequence and evaluate the H_0 innovation; (2) record alarm time, refine physical onset candidates, and fit a common pre-event state; (3) score every admissible physical family/location with its own event operator and explicit numerical status; (4) retain the frozen within-family quotas; (5) apply the same staged nonlinear budgets to all survivors while checkpointing objective and stationarity; (6) finalize every remaining candidate to the same strict gate; and (7) compare only validly converged modes with consistent priors, normalizers, and boundary treatment. Outputs before step (6) are candidate sets and diagnostics, not final source decisions. This route is retrospective whenever onset refinement or smoothing uses samples after the decision time; it has no measured end-to-end latency guarantee.

Several discarded results in the dossier were caused by implementation changes rather than by power-system information limits. A generator screen based only on the algebraic Jacobian missed its differential pathway; a previous joint update computed but did not apply the severity step; and an outage candidate was once switched at frame zero instead of the inferred onset. A full-window H_0 fit absorbed an event into the null reference, while independent caps on state and severity steps invalidated a quadratic proposal score. Unscored fault candidates inherited catalog order through a tied sentinel value. One detection-delay calculation used the window start rather than its end, and some failed integrator returns were encoded as nonfinite values despite finite unfinished states. The corrected pilot treats each as a separate numerical or semantic gate. Its four-case truth survival must not be merged with superseded rankings from those earlier variants.

D. EVIDENCE AND THE FINAL COMPARISON GATE

For converged, regular, identifiable interior modes only, the local Laplace approximation is

$$\log p(Y \mid H) \simeq -J_H(\widehat{u}_H) + \tfrac{d_H}{2}\log(2\pi) - \tfrac{1}{2}\log|\nabla^2 J_H(\widehat{u}_H)|, \tag{170}$$

where u_H uses a common feasible measure and J_H includes all model-dependent prior and likelihood normalizers. The normalizer, prior scale, curvature determinant, and any boundary or quotient correction must be retained when comparing families of different dimension. Section E-F derives the boundary-aware alternative. A candidate that stops at $B10$ or violates the strict stationarity/solver gate cannot be made comparable merely by evaluating (170). The pilot has not demonstrated exact normalization against the load-bank restriction $\mathcal{R}_0$.

E. DOSSIER-REPORTED NUMERICAL LEDGER

Table 18 preserves the numerical chronology without treating small pilot denominators as population rates. "Dossier AV" means the PDF claims an external campaign package supports the entry; it does not mean this repository has reopened that package. "TR" is the dossier's own terminal-report tier.

The case-specific terminal report places converged LOAD@27 and LOAD@26 ahead of true LOAD@28. It quotes approximate objectives -29986.9, -29985.9, and -29980.7, respectively, with evidence values around 4609.98, 4610.22, and 4607.17. Those values are deliberately absent from the main manuscript: their sign convention and normalizing terms must be checked against the final archive before quantitative model comparison. No exact generator

margin was supplied. The fault and line comparisons are not physically interpretable until both truth and nearest competitors pass a common strict convergence gate.

F. REQUIRED CLOSURE BEFORE A MULTI-FAMILY CLAIM

The next frozen audit must first replay each event operator, measurement conversion, onset, duration, and initial state under the same truth and inference model. It must then use an independent finalizer for the fault/line candidates, retaining every failed solve in the denominator. For converged load/generator mislocalizations, successive oracles should fix the true pre-event state, timing, severity, and trajectory, then remove noise and add a counterfactual nearby PMU. Only a persistent small whitened truth-to-competitor margin after those controls supports an information-limited explanation. The final-survivor archive must supply per-candidate status, objective, prior and likelihood constants, Hessian measure, and evidence normalization. A larger source-disjoint holdout, operational normal exposure, and the $\mathcal{R}_0$ reduction test are still needed. The current pilot cannot be used to upgrade the separate $\mathcal{E}_{\mathrm{B}}$ load-support result into a validated multi-family joint estimator.

APPENDIX I RESEARCH DECISIONS BEHIND THE MAIN NARRATIVE

The article's main text now follows one question: which physical intervention explains a sparse PMU history, and when is that explanation distinguishable? The supporting studies below retain different estimators and denominators. They are evidence for particular steps in that argument, not a pooled test of a completed joint estimator.

A. THE CAUSAL IMPLEMENTATION'S LOCAL AMPLITUDE INTEGRAL

This calculation describes the retained local Gaussian history model used by R11, not a replacement for the batch estimator's GH31 quadrature. For one load-history interval, write its new-report residual as

$$e = sa + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, S), \qquad a \sim \mathrm{TN}_{[l,u]}(\mu, v).$$

The physical transition/observation operators supply s; conditional state and colored-noise propagation supply S. Here μ, v parameterize the underlying untruncated normal density, not its bounded mean and variance. Define

$$g = s^\top S^{-1} s, \quad q = s^\top S^{-1} e, \quad v' = \frac{v}{1+vg}, \quad \mu' = \mu + v'(q - \mu g),$$

and $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Completing the square gives

$$\begin{aligned} \log p(e \mid h) = & -\tfrac{1}{2}(e^\top S^{-1} e + \log|S| + n_y \log(2\pi)) \\ & + \tfrac{1}{2}\left[\log(v'/v) + \mu'^2/v' - \mu^2/v\right] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \end{aligned}$$

Reported amplitude moments use the corresponding truncated-normal moments. The integral is evaluated over the retained amplitude intervals; it is not replaced by the best-fitting amplitude alone. Fixed line-mode hypotheses use their switched operator without a load-amplitude integral.

The square completion is standard. Its role here is to show precisely how a physical sensitivity and an uncertain intensity enter the executable likelihood. It is exact for this conditional Gaussian model; linearization, covariance sharing, nuisance substitution and history pruning prevent an exact unrestricted nonlinear-posterior claim. The frozen implementation is `integrated_loglike` in the R11 `observer/static_rx.py`; the one-report model weights call it before current-report screening. Rolling proposals do not create a second copy of the same observations.

B. STATE RECOVERY AND MODEL ADEQUACY

The matched B2 mean hidden-voltage error is 0.009634%, compared with 0.3402% for the equilibrium reference. Under the tested operating shifts, frozen errors of 0.19885%, 0.38834% and 0.57795% become 0.00525%, 0.00749% and 0.00974% after regularized physical recentering. The nuisance information rank is 25 of 43. This supports a useful point target while leaving parameter non-uniqueness and joint covariance failure visible.

C. BATCH SOURCE RESOLUTION

The implemented batch estimator retains GH31 amplitude integration, W2 covariance, a fixed operating point and known onset/family. It evaluates the complete zero-, one- and two-source catalog; it does not supply the causal R11 decisions.

D. WHAT THE REFERENCE ESTIMATOR STILL DOES NOT CLAIM

The historical supervised bank spans 690 ANDES trajectories, with detection F1 0.989 and physical Top-1 0.592 on its own ordinary split. Its held-source stress test gives 3/303 exact decisions. These are different populations from H04/C01, so their scores cannot be subtracted to claim an improvement of the hackathon leaderboard. The unrestricted joint state/event/integrity estimator and learned discrepancy layer remain proposals. The pre-revision manuscript and its complete original main-text derivations are preserved in `archive/narrative_before_20261030/`.

TABLE 19. Exploration changes the next inference question, not the test labels.

Question explored	Observation	Decision retained	Boundary
Can eight PMUs constrain hidden voltage?	Accurate matched reconstruction; shift degrades it	Keep the physical state and operating context explicit	State recovery is not source attribution
Can an offset repair all mismatch?	Hidden-state recovery needs AC-feasible recentering	Separate physical recentering from measured residual centering	R11 intercept is not a new hidden-state solution
Does a source label transfer?	Legacy held-source test fails	Generate physical hypotheses for candidate assets	Causal unseen-source transfer remains untested
Does detecting multiplicity locate both sources?	Weak exact supports overlap	Profile severity/nuisance and keep alternatives	Batch onset/family known
Does every plausible multi-family proposal become a correct final decision?	The exploratory pilot does not establish that	Keep E_J as proposed, not a validated replacement for E_B	Dossier-only results remain secondary
Can a frozen physical bank follow successive events?	H04 separates timely recognition from eventual attribution	Retain dynamic recovery and compare nominal/mixture	Development, known catalog
Can stricter episode admission remove extras for free?	It rejects a real BUS16 intervention	Keep the failed ablation and freeze co-references	Zero extras is not uniform improvement
Does the result persist under an independent draw?	C01 is prespecified	No further method selection on that examination	Results enter only after complete evaluation

TABLE 20. Fresh prospective 137-hypothesis confirmation. The bank contains H_0, 16 single-source hypotheses, and all 120 two-source supports. Repeated noise realizations are decisions, not independent physical trajectories. Top-k is exact support recovery and is undefined for H_0 and single-source rows in this exported support table.

Regime	Physical trajectories	Decisions	Cardinality (%)	Top-1 (%)	Top-3 (%)	Mean $p(K = 2)$
No event	—	100	100.0	—	—	0.000
Single source	128	384	93.8	—	—	0.051
Weak–weak pair	1920	5760	37.4	24.4	59.8	0.427
Weak–strong pair	480	1440	77.6	66.2	95.8	0.808
Moderate pair	480	1440	99.7	98.8	100.0	0.996
Finite pair	480	1440	100.0	100.0	100.0	1.000

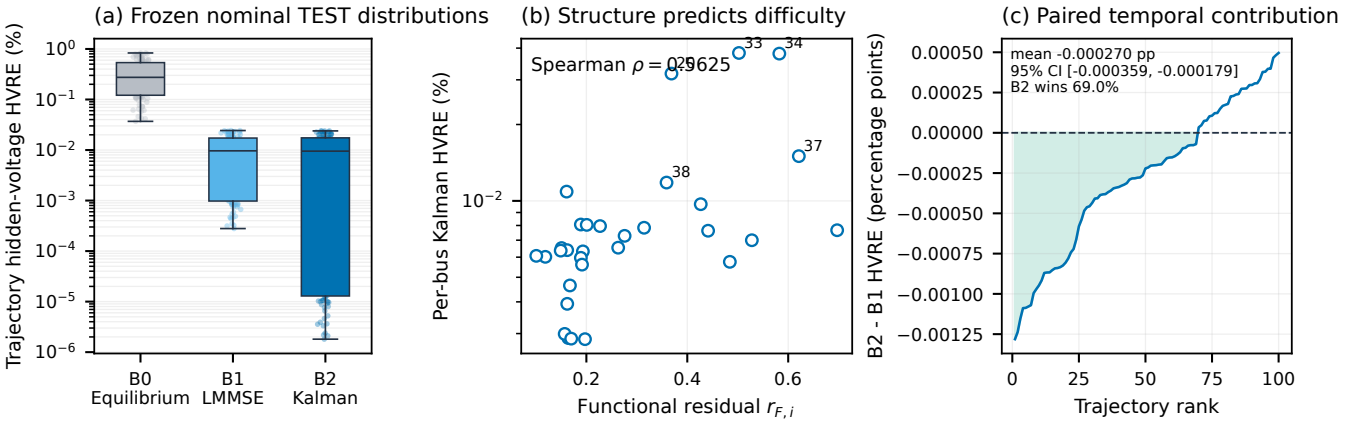


FIGURE 4. **Expanded matched state evidence, retained from the original frozen study. It evaluates hidden-voltage reconstruction and does not certify source identity.**

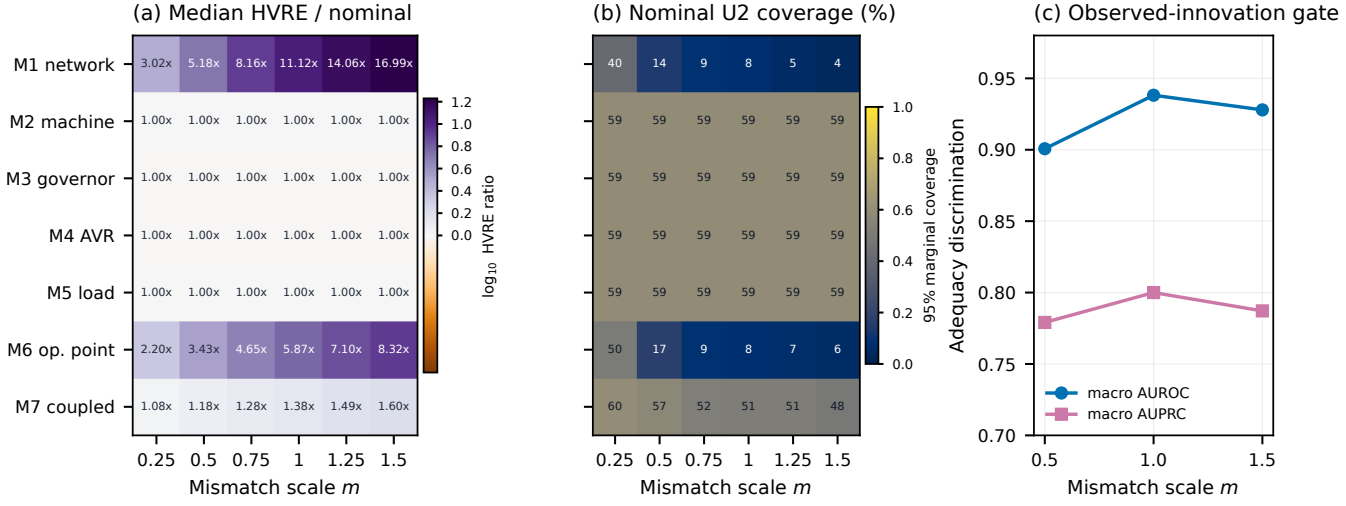


FIGURE 5. **Expanded model-mismatch evidence. Physical and operating shifts are distinct from instrumental covariance corrections.**

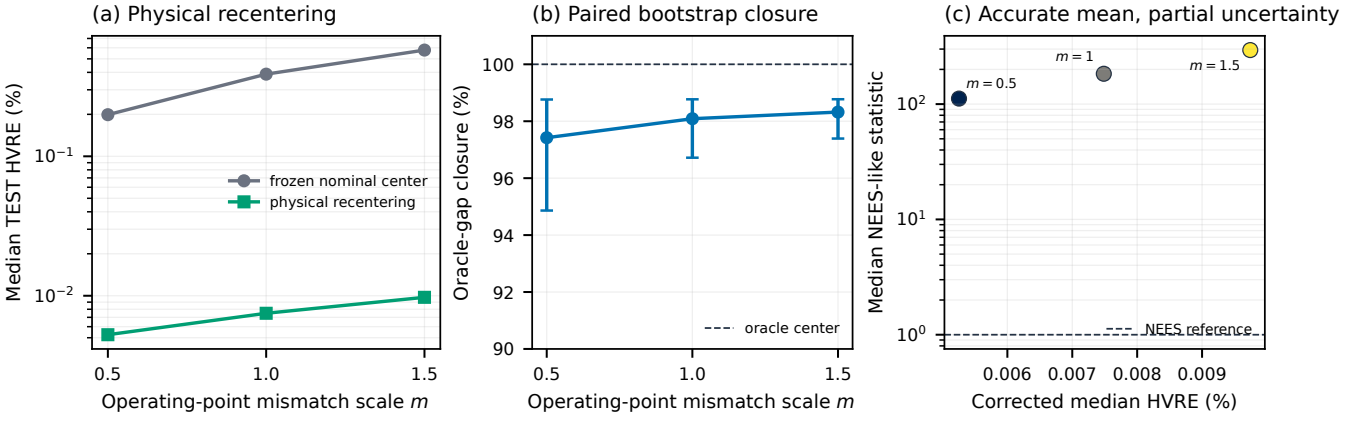


FIGURE 6. **Physical recentering repairs much of the tested point-estimate error while the joint uncertainty diagnostic remains inadequate.**

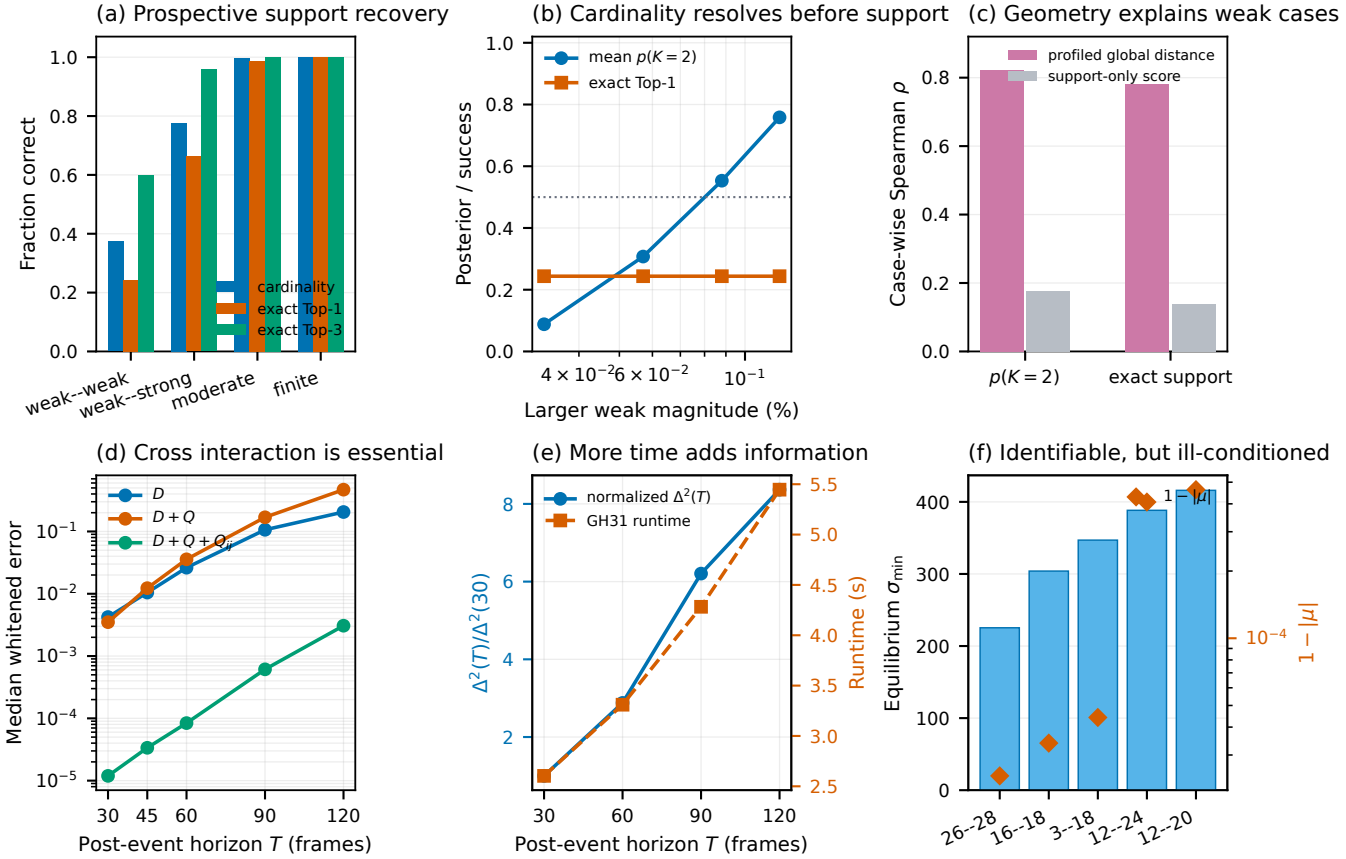


FIGURE 7. Expanded source-resolution study. Prospective recovery, retrospective profiled geometry and the longer-horizon diagnostic have distinct roles; the latter panels do not replace a new prospective test.

REFERENCES

[1] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.
[2] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023, posted June 16, 2023; cited as a preprint.
[3] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.
[4] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.
[5] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.
[6] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.
[7] B. Qu, Z. Wang, B. Shen, H. Dong, and H. Liu, "Event-based joint state and unknown input estimation for energy networks: Handling multi-machine power grids," *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 1, pp. 253–264, 2023.
[8] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, J. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Applied Energy*, vol. 358, p. 122602, 2024.
[9] H. Yue, W. Zhang, U. C. Yilmaz, T. Yildiz, H. Huang, H. Liu, Y. Lin, and A. Abur, "Graph-learning-assisted state estimation using sparse heterogeneous measurements," *Electric Power Systems Research*, vol. 235, p. 110644, 2024.
[10] S. Moshtagh, B. Azimian, M. Golgol, and A. Pal, "Topology-aware graph neural network-based state estimation for PMU-unobservable power systems," *IEEE Transactions on Power Systems*, 2025.
[11] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
[12] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
[13] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
[14] A. Kummerow, M. Dirbas, C. Monsalve, and P. Bretschneider, "Siamese sigmoid networks for the open classification of grid disturbances in power transmission systems," *IET Smart Grid*, vol. 6, no. 2, pp. 136–146, 2023.
[15] Y. Hu and Z. Cheng, "Generative AI for open-set event detection and classification in power systems using synchrophasor data," *IEEE Transactions on Industrial Informatics*, 2026, early access.
[16] M. Rimkus, P. Kokoszka, D. Duan, X. Wang, and H. Wang, "Graph neural networks for the localization of faults in a partially observed regional transmission system," *Scandinavian Journal of Statistics*, vol. 52, no. 2, pp. 572–594, 2025.
[17] Z.-M. Zhai, M. Moradi, and Y.-C. Lai, "Detecting attacks and estimating states of power grids from partial observations with machine learning," *PRX Energy*, vol. 4, no. 1, p. 013003, 2025.
[18] J. L. Mayorga Taborda, Y. Africano Rodriguez, and F. Jimenez, "Benchmarking dynamic frequency estimators for low-inertia IBR grids: A latency–robustness trade-off analysis," in *2026 5th International Conference on Smart Grid Synchronized Measurements and Analytics (SGSMA)*. IEEE, 2026.
[19] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Englewood Cliffs, NJ, USA: Prentice Hall, 1993.
[20] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.
[21] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ, USA: Wiley-Interscience, 2006.
[22] A. B. Tsybakov, *Introduction to Nonparametric Estimation*. New York, NY, USA: Springer, 2009.
[23] A. Rotnitzky, D. R. Cox, M. Bottai, and J. M. Robins, "Likelihood-based inference with singular information matrix," *Bernoulli*, vol. 6, no. 2, pp. 243–284, 2000.
[24] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.
[25] A. M. Stuart, "Inverse problems: A bayesian perspective," *Acta Numerica*, vol. 19, pp. 451–559, 2010.
[26] H. E. Rauch, F. Tung, and C. T. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, vol. 3, no. 8, pp. 1445–1450, 1965.
[27] B. M. Bell, "The iterated kalman smoother as a Gauss–Newton method," *SIAM Journal on Optimization*, vol. 4, no. 3, pp. 626–636, 1994.
[28] S. Särkkä and L. Svensson, "Levenberg–marquardt and line-search extended kalman smoothers," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 5875–5879.